Investify

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October, 2026

Students' Submission

Unsigned working draft. The submission statement and signatures require actual student review/confirmation before submission. The preparation date is not a confirmed assessment date.

We certify that this report is submitted as the FYP-1 mid-report for the project titled *Investify*.

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Anti-Plagiarism Statement

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The students certify that this report is their own work and that all sources used in preparing the report have been appropriately acknowledged and referenced.

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We declare that the work documented in this report represents the work completed by the project team up to the FYP-1 mid-report stage. Contributions, tools, external components, and relevant sources are disclosed where applicable.

Abstract

Investify is an educational equity decision-support prototype that combines technical indicators, news sentiment and explanations checked against recorded evidence. Its verifier checks visible prose and JSON independently and supplies a deterministic fallback. A prototype risk questionnaire conditions a separate retrospective linear comparator.

A fresh NVDA experiment processes 1,509 price sessions and 6,768 revision-conservative news rows. The clean five-day model obtains macro-F1 0.2843 versus a stratified-random baseline of 0.3287. A deliberately centred indicator triggers the leakage canary. Both corrected calibration modes produce only Buy decisions on 186 evaluation sessions; improved Brier scores therefore do not establish useful classification.

Original CryptoBERT and Cardiff classifiers complete the same 24 frozen MSFT posts, scoring macro-F1 agreement of 0.2564 and 0.6016 against AI reference labels. The small, imbalanced reference has no independent human validation, and deployment selection remains open.

Recorded-model, calibration and native TreeSHAP replay pass. The interfaces expose evidence, failure cases and recorded comparisons. Six assistant-teacher explanations form a construction pilot; the planned Phi-3-mini student remains untrained. Broader generalisation, live LLM usefulness, agreed risk semantics, institutional completion and genuine team review remain outstanding.

Acknowledgments

The team will finalize acknowledgments to reflect actual guidance, review and contributions. Supervisor identity follows the supplied project record. No additional participant, reviewer or contributor is invented in this working draft.

Executive Summary

Investify separates educational equity predictions, recorded evidence, explanations and checking. A label alone does not communicate its inputs, uncertainty or limitations. The implementation exposes five technical and six news features, checks visible prose independently of JSON, rejects known contradictions and supplies a verified deterministic fallback. Prototype risk answers condition a separate retrospective linear blend; they do not personalise the saved classifier.

The preserved September experiments and 372 historical decision records remain unchanged. A fresh Alpaca experiment now restores a genuine input-to-model execution path on 1,509 NVDA price sessions and 6,768 revision-conservative news rows. Original FinBERT scores 5,748 unique headlines offline. Six leakage arms run at one- and five-day horizons with fixed model settings, shared row sets and six folds. The clean five-day macro-F1 is 0.2843 against a training-distribution random baseline of 0.3287; the centred-feature mutation has lift +0.2104. The weak clean result and deliberate failure are reported together.

Corrected frozen-classifier and refit-transfer calibration modes each emit 186 Buy decisions. Raw frozen decisions are Buy 40, Hold 51 and Sell 95; their fitted isotonic mappings make Buy largest on every evaluation row. Frozen Brier improves from 0.7929 to 0.6105 while macro-F1 falls from 0.3182 to 0.2238. A fixed comparator estimated only from the 181 available calibration labels has evaluation Brier 0.6095. Improved Brier against the raw model therefore does not demonstrate an advantage over a simple class-prior forecast. Model-byte, scaler, margin/probability, calibration, label-endpoint and native TreeSHAP replay pass. All 372 new complete-guided explanations verify separately from the historical records.

The social pilot retains 200 real MSFT posts, 117 usable AI references and 83 disclosed exclusions, with 93 development and 24 chronological held-out posts. Both original classifiers now complete the same 24 held-out posts without errors. CryptoBERT predicts only neutral and has macro-F1 agreement 0.2564; Cardiff has 0.6016. Exact agreement is 15/24 and 16/24. Cardiff is the exploratory benchmark leader, with only two negative reference examples. Publisher bytes, tokenizer/configuration identities and lossless original Cardiff serialization are verified. Its unused native pooler is retained with exact

classifier-logit parity. Independent human validation and deployment selection remain open.

The rebuilt interface renders one workspace and one of four walkthrough steps at a time. The current fourteen-case AppTest scope exercises real selection, evidence, profile, corruption/fallback and stale-result/refusal behavior. Its Experiments workspace recomputes receipt-bound social scores without inference. The earlier 28-check mounted-tab scope is historical. Other dated scopes retain 179 scientific contract passes with six legacy-input skips, eight admission-refusal checks, original-model loader/pooler fixtures and separately executed portable HTTP checks. These scopes overlap and are not added into one full-suite total. Browser/projector and genuine participant review remain distinct; prepared remote CI has not executed.

The current assistant teacher construction pilot contains six verified explanations and four training/two validation exports. Factual JSON supervision is assembled from original records; the exact serving revision is unavailable. Phi-3-mini remains the planned, untrained student, and Mistral is fallback. QLoRA belongs to the later milestone. No measured human-comprehension or training-efficacy claim follows from this pilot.

Remaining first-mid work includes final report/deck delivery and inspection, required institutional facts and applicable signatures, genuine member contributions and review/rehearsal, current repository evidence and agreed social/risk acceptance criteria. Broader prediction, social integration into the engine, live LLM evaluation, personalised modelling and deployment remain distinct later work. This report is an unsigned working draft.

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List of Abbreviations

FYP	Final Year Project
FR	Functional Requirement
NFR	Non-Functional Requirement
API	Application Programming Interface
CI	Continuous Integration
CD	Continuous Delivery / Deployment
CV	Cross-Validation
F1	Harmonic Mean of Precision and Recall
FNSPID	Financial News and Stock Price Integration Dataset
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
MACD	Moving Average Convergence Divergence
NYSE	New York Stock Exchange
OBV	On-Balance Volume
PoC	Proof of Concept
QA	Quality Assurance
QLoRA	Quantized Low-Rank Adaptation
RSI	Relative Strength Index
SHAP	Shapley Additive Explanations
SMA	Simple Moving Average
UTC	Coordinated Universal Time

Chapter 1

Introduction and Project Vision

1.1 Project Overview

Investify (F26-003-Investify) is an educational investment advisory project for retail investors. The planned system combines technical indicators, news and social sentiment, and risk information to present a one-week Buy, Hold, or Sell view with understandable, checkable evidence. The proposal was approved with minor modifications concerning contribution clarity, the language-model role, evaluation, transparency, and domain understanding.

1.2 Problem Statement

A retail investor evaluating a company must interpret historical price indicators and a large volume of news and social commentary. A decision label alone does not explain which signals influenced it, how conflicting evidence was handled, how uncertain it is, or whether an action fits the investor's risk constraints. A fluent explanation can misstate recorded values or attributions, while an evaluation using information unavailable at decision time can overstate model quality.

Investify addresses these issues through a temporally controlled evaluation pipeline and explanations tied to recorded inputs and model attributions. Its outputs are educational and advisory; predictive usefulness must be established against explicit comparators, and the system does not execute transactions.

1.3 Problem Context and Motivation

The baseline contains a historical NVDA experiment using technical indicators and financial-news sentiment. It found no demonstrated predictive signal, and its calibrated evaluation records all select Buy. Explaining those records makes the computation and its limitations inspectable; it does not establish a useful trading signal. Financial-text classification supplies evidence about text tone rather than the truth of a headline or a promised return [1].

1.4 Project Vision

The user can inspect a view's evidence, understand its uncertainty and coverage, and ask questions without receiving invented values or unsupported causes. The LLM communicates the recorded decision. A separate claim verifier and deterministic fallback govern supported factual statements.

1.5 Project Objectives

Objectives cover point-in-time alignment, engine/baseline comparison, equity social-model transfer and news/social disagreement, capacity/tolerance reconciliation, verified explanations and user understanding, and a reproducible measured system. Implemented components, measured findings, and planned objectives are reported separately.

1.6 Project Scope

The current prototype covers preserved NVDA artifacts plus a freshly executed Alpaca news/price experiment at one- and five-session horizons. A separate original-classifier selection study uses frozen MSFT social posts. Full scope targets a small equity universe, a one-week user horizon, social sentiment, genuine risk conditioning, and a verified LLM. Price-only continuously traded assets provide a later calendar experiment. Brokerage integration and trading execution are excluded.

1.7 Project Modules

Modules cover ingestion/storage, calendar/as-of alignment, feature/sentiment construction, evaluation/controls, prediction/calibration/attribution, risk profiling, explanations/verification, and interface/deployment. The new increment adds complete evidence context at the explanation/interface boundary.

1.8 Stakeholders and User Classes

The primary user seeks educational analysis. The operator maintains data/runtime health. The supervisor evaluates scope and evidence. Each team member must explain and modify their actual work and disclose external assistance and components.

1.9 Success Criteria

Criteria include causal alignment checks, injected faults, held-out predictive/calibration metrics, explanation measures, interpretation tasks, and runtime/installability checks. Existing explanation targets are numeric fidelity at least 0.98, directional consistency at least 0.95, recall at least 0.80, and unverifiable rate at most 0.15. Planned bounds are not measurements.

1.10 Constraints and Assumptions

The original September training store/raw corpora remain unavailable, and fresh upstream changes are unverified. A new Alpaca scientific run and the original social-candidate comparison now execute with separate identities. Live LLM/student inference and comprehension evaluation remain pending. Six recorded teacher examples demonstrate construction under a bounded contract. The prototype questionnaire affects a separate technical/news comparator; it does not condition the saved XGBoost predictions or implement the proposal's validated risk instrument. Context/verification checks can run without the training stack.

1.11 Ethical, Legal and DEI Considerations

The interface must explain data age, coverage, uncertainty, actual profile effect, and failure states. An AI explanation is not a guaranteed prediction [2]. Personal data, annotations, and acceptance work require appropriate privacy, licensing, consent, and truthful records. The advisory boundary must remain clear throughout the journey.

1.12 Stakeholder Goals and Educational Opportunity

Role	Need/problem	Current boundary
Retail learner/reviewer	Understand a recorded class, its input values and uncertainty.	Historical educational case; no useful live trading signal or investor suitability claim.
Supervisor	Trace requirements/design to a working module and discuss a failure.	Live checking/profile arithmetic and honest recorded research; actual baseline/action approval requires review.
Maintainer/team	Reproduce a case and identify changed or unavailable evidence.	Explicit source registries, digests, retained receipts and local setup instructions.
Research evaluator	Distinguish model quality from explanation fidelity.	Separate chronological/scientific and bounded-explanation results; human comprehension remains unmeasured.

The opportunity is educational decision transparency: combining maintained indicator meanings with observable checking of a recorded explanation. No customer demand, business adoption or revenue evidence has been collected. The intended user roles are specified, but stakeholder consultation and a named real client/approver must be recorded by the team; no invented external client is inserted.

1.12.1 DEI and Ethical/Legal Design Implications

The interface exposes dates, missing evidence and units, supports keyboard controls and uses text verdicts alongside color. Prototype age/capacity weights need fairness review.

Chapter 2

Research and Existing Solutions

2.1 Research Method / Approach

This preliminary review uses primary research abstracts, the actual sentiment checkpoint's model card, official exchange/API documentation, and the preserved project artifacts. The comparison is focused on sentiment inputs, time alignment, attribution and explanation consistency. It is not a completed systematic literature review or evidence that no other product provides similar features.

2.2 Existing Solutions / Products

Araci's FinBERT work investigates domain-adapted language models for financial sentiment [3]. The implemented checkpoint, `ProsusAI/finbert`, outputs positive/neutral/negative sentiment scores and is documented separately by its provider [1]. Investify transforms these outputs into historical news features; their availability alone does not establish return predictability or social-domain transfer.

FNSPID combines financial news with stock-price data for forecasting research [4]. The project's preserved subset probe concerns the actual selected files/tickers and timestamp precision. It must not be generalised into a claim that every record in the published dataset has the same limitation. Supplemental Alpaca/Benzinga results remain a separate experiment.

Tree-based SHAP work provides consistent local feature attributions [5]. Investify uses stored predicted-class attributions to explain raw model scores. The verifier and educational display are downstream additions, not inventions of TreeSHAP or guarantees of economic causation.

Parcalabescu and Frank distinguish output-level self-consistency tests from stronger

claims about a language model's internal explanation faithfulness [6]. This motivates a precise boundary: Investify checks supported statements against an external decision record and a maintained glossary. It does not prove all natural-language meaning or the LLM's internal reasoning faithful.

2.3 Comparison of Existing Solutions

Work/component	Relevant capability	Investify's use or evaluation question
FinBERT/checkpoint	Financial-text sentiment classification	Historical news features; equity social transfer still needs a labelled study
FNSPID	Integrated news/price research data	Audit subset coverage, publication precision and calendar alignment
TreeSHAP	Local attribution for tree models	Validate additivity and preserve raw-score units in explanations
Natural-language consistency research	Limits of explanation faithfulness tests	Define record-consistency claims and test concrete corruptions
Numeric deterministic template	Reproducible recorded facts	Initial explanation comparator and verified fallback
Definition-enhanced template	Same facts with maintained feature meanings	Fair comparator for any claimed LLM comprehension benefit
Local constrained LLM pilot	Structured candidate generation	Adapter tested; actual model outputs/evaluation unavailable

2.4 Limitations and Gaps

The engineering gap demonstrated locally is that a correct JSON payload could hide an incorrect displayed explanation. Concrete regressions establish this defect and its correction. A broader research novelty claim requires comparison with the closest record-grounded generation and checking approaches. Adding an LLM or SHAP alone is insufficient. Human understanding, unrestricted language verification, social transfer and useful

stock prediction remain unestablished.

2.5 Closest Factuality and Annotation Work

FActScore decomposes generated content into atomic facts and assesses support against a knowledge source, including a retrieval/model-based estimator [7]. Its open-domain factual-precision setting motivates claim-level accounting. Investify instead limits accepted language to a maintained grammar and external numerical record, and checks required visible evidence separately from factual precision. Accurate included facts do not establish that required evidence was included. This is a scoped engineering distinction; a new general factuality method or empirical advantage over FActScore is not demonstrated.

RARR researches attribution and revises unsupported generated content while preserving the original output as much as possible [8]. Investify's correction policy rejects a failed bounded-contract attempt, optionally retries, and emits a checked deterministic guide. The planned pilot supplies that complete guide to the model. A meaningful educational advantage therefore requires further evaluation beyond successful reference reproduction.

Gilardi and colleagues study ChatGPT text annotation on relevance, topic, stance and framing tasks [9]. That research motivates investigating AI references, but does not validate the current financial labels, unknown exact serving revision, subjective exclusions or historical MSFT cohort. Candidate studies must report agreement with one AI reference pass and retain the independent-human-validation limitation.

2.6 Official Publisher Model Policy

The current assistant is the active explanation teacher under user direction; Phi-3-mini remains the planned original Microsoft student, with its exact checkpoint pending. Mistral is a fallback and has not been called. Six controlled-prose teacher examples are recorded, with factual JSON assembled from the original records as disclosed hybrid supervision. Exact teacher serving revision, student inference, training benefit and user comprehension remain unmeasured.

Local model baselines must use original publisher weights or publisher-issued quantizations, excluding third-party fine-tunes and behaviour-modified variants. Qwen2.5-7B-Instruct-GGUF and Qwen3-8B-GGUF are earlier inactive optional local configurations, not the selected teacher or student. Their catalog declares exact revisions and every weight-shard SHA-256/size; actual weight verification is required before those optional studies. Acquisition was refused and no replacement inference score exists. Frozen AI

references and historical classifier artifacts remain separate. The project's later QLoRA adapter must be versioned and disclosed separately from unchanged student weights.

2.7 Proposed Solution

Investify combines temporal evaluation controls with explanation claims traceable to recorded inputs/attributions. Visible text and JSON are checked independently, known contradictions fail, and unsupported/failed generations select a verified deterministic explanation. The local pilot includes exact maintained definitions and compares numeric template, definition-enhanced template and real generations on a frozen chronological selection. No quality score is reported if the provider returns no completion.

2.8 Complex Computing Problem

The problem combines asynchronous text and trading calendars, overlapping prediction horizons, fold-local processing, model calibration/attribution, and generation whose displayed claims must remain consistent with external numerical facts. The interfaces have competing constraints: understandable explanations need adequate coverage, while checking must avoid accepting unsupported claims or rejecting all useful content. Institutional mapping to the required complex-computing-problem criteria still requires supervisor/team review.

2.9 Technical Challenges

Publication precision, session/half-day boundaries, missing data, calibration collapse and raw-score interpretation complicate the baseline. Original September inputs remain absent; separate genuine Alpaca inputs and a scientific Python 3.11 runtime now support executed experiments. Fresh default LM Studio/Ollama probes return no model identifier. The inspected model directory names an old C: target, while E: contains application/backends and a bundled embedding model but no discovered generative weights. Global configuration is unchanged; these observations do not establish all failure causes or generation/deployment completion.

The social-model study distinguishes post sentiment from return prediction. CryptoBERT's provider documents crypto-social training, bearish/neutral/bullish class indices and a 128-token training length [10]; a generic Twitter alternative documents different sentiment names and mention/URL preprocessing [11]. Equity-domain suitability therefore needs the actual labelled study, not a model-name preference. The local-only run-

ner records explicit mappings, token/truncation information and failed outcomes; missing/randomly initialized classifier heads are refused. Earlier tests use fixture backends; the later original-publisher comparison now executes both classifiers with raw-logit and byte-bound receipts.

FinLin provides a multi-source entity-level research design, with individual annotations in its described schema [12, 13]. The inspected repository contains descriptions/figures rather than the corpus; no access request has been sent. The comprehensive StockTwits project describes public-access joined message/metadata tables and user sentiment tags [14]. Such tags do not provide new independent human labels or a neutral class. Corpus access, permitted use, published-score handling and pretraining overlap require explicit review. A chronological benchmark split alone does not prove absence from model pretraining.

2.10 Research Findings and Design Decisions

Stocks represent ownership interests, and price movement can reflect company-specific or wider market events [15]. The prototype's technical/news features do not provide a full valuation or represent every fundamental/macroeconomic driver. A worked explanation must distinguish what its actual inputs measure from unsupported claims about why a market price changed. Corporate-action adjustment, broader drivers and changing regimes still require dedicated review.

NVIDIA's official 2021 FAQ documents a four-for-one split and trading on the new basis from 20 July [16]. As a worked arithmetic illustration, one share at 100 units becoming four shares at 25 units preserves 100 units of value; treating the per-share change as a 75% investment loss would omit the share-count change. These numbers illustrate arithmetic and are not observed NVDA prices. Yahoo describes adjusted close as accounting for splits and distributions [17]. The current ingest disables yfinance's client adjustment and omits action columns [18]; that setting alone does not verify provider Close/Volume are as-traded. The raw snapshot, action history and volume basis require audit before any adjustment or performance claim. This issue has not been established as a cause of constant-Buy output.

Official exchange hours include regular and early-close schedules [19]; alignment must therefore use an explicit decision-time contract and exchange calendar rather than calendar-date equality. The calendar cutoff/proposal reconciliation remains open. Model-card sentiment outputs motivate a news-specific pipeline and separate equity-social study. Attribution theory motivates additivity checks and clear raw-score units. Explanation-consistency literature motivates bounded, explicit claims rather than a claim of universal semantic faithfulness. Local structured-output documentation supports JSON-schema

requests, but schema conformance does not replace factual checks [20].

2.11 Actual AI Reference Pilot

The author-released StockNet corpus provides original social-post content, identifiers and timestamps [21, 22]. The first NVDA sampling declaration could not execute because that ticker is absent from the released directory list. MSFT was declared before content acquisition. The actual AI reference pilot contains 200 real MSFT tweets: 117 usable labels (31 positive, 67 neutral, 19 negative) and 83 explicit exclusions. Its frozen split is 93 development/24 held out, with only two negative held-out references. One session pass is recorded with full prompt/input/raw-output/timing and declared model-identity provenance; agreement/kappa are null and independent human validation is absent. Both original candidates now complete the same 24 held-out posts; their exploratory agreement results are reported in Chapter 8. Public historical posts may occur in annotation or candidate pretraining. Exact deduplication leaves near-duplicates and retweets. Whole-post sentiment candidates and target-equity references differ in scope. These limits prevent an independently human-validated accuracy or NVDA transfer claim. The retained MIT repository notice does not authenticate third-party tweet rights; this workspace has not publicly redistributed the corpus.

Chapter 3

Proposal Feedback and Iteration Plan

3.1 Proposal Feedback Summary

Project F26-003-Investify was approved with minor modifications. The panel requested clearer contributions beyond leakage, the LLM's user benefit, technical/sentiment explanations, refined problem/model descriptions, explicit metrics, an improved PoC, user transparency, and deeper market understanding.

3.2 Proposal Action Register

The working register preserves the full supplied wording. These identifiers organize project actions; owners and dates require team agreement.

ID	Required response	Current evidence/status
PF-01	Clarify contribution	Revised statement drafted; related-work comparison pending.
PF-02	Explain LLM benefit	Six actual recorded teacher examples; live inference and demonstrated user benefit pending.
PF-03	Explain technical/sentiment evidence	Complete technical/news context and recorded social comparison; live LLM benefit pending.
PF-04	Refine problem statement	Working revision drafted.

PF-05	Clarify pretrained models	Original classifiers compared; deployment criteria and live LLM evaluation pending.
PF-06	Identify metrics	Matrix and current checks documented.
PF-07	Improve PoC beyond leakage	Corrected verifier and browser journey checked; supervisor demonstration pending.
PF-08	Transparency throughout journey	Data, coverage and risk limits exposed; current failure/fallback paths tested; human understanding pending.
PF-09	Market understanding	Referenced notes begun; broader research pending.

3.3 Iteration Strategy

Each increment supplies a working path, tests, and evidence. The independent code copy retains Git history and uses base commit 1dc24e0. Existing documents and the original checkout remain protected. Upstream access failure is recorded rather than taken to mean no new work exists.

3.4 Iteration 1 Plan

Original commitments cover ingestion, calendar mapping, indicators, news sentiment, social-model selection, the linear baseline, leakage tests/mutation detection, an interface slice, and requirements/design. The first new increment establishes artifact validation and complete evidence presentation. Next work connects and evaluates a real prompted explainer, then addresses failure behavior and risk semantics.

3.5 Iteration 2 Plan

Later work follows the proposal's engine, divergence, deployment, grammar, language-model comparison, runtime, and acceptance milestones. The feedback does not itself require the entire QLoRA programme at this mid.

3.6 Work Breakdown and Responsibilities

The planned work division follows slide 13 of the preserved proposal, including its SmartArt parent/child assignments. The table provides the original planned responsibilities for implementation and defence preparation. The current student-record spelling is Soban Ahmad; the source slide spells Soban Ahmed.

Member	Proposal-assigned responsibilities	Completion evidence
Ahmed Javaid	point-in-time harness; leakage tests; dual-timestamp schema; parity test; deployment	Actual member work and defence to be recorded.
Rafay Ul Haq	risk profiling; sentiment pipeline; degraded operation; acceptance testing; frontend	Actual member work and defence to be recorded.
Soban Ahmad	decision engine; Calibration; SHAP; Verifier; QLoRA distillation	Actual member work and defence to be recorded.

The proposal assigns the harness to Ahmed, while historical repository records associate baseline work with Soban. Planned responsibility and actual past authorship are different evidence. The team must record actual implementation, review, tests understood and assistance used; historical contributions remain preserved. QLoRA distillation remains later work.

3.7 Progress Against Plan

The earlier 12 October selected suite has 212 passes and no skips, including 20 actual Streamlit cases. All 372 records/complete-guided templates, 62 live HTTP checks and 228 separate comparator arithmetic probes validate. A prototype profile/blend, restored viewer, social-study validator/evaluator/view and offline classifier runner are implemented. Six actual assistant-teacher explanations pass the bounded verifier; this does not establish added explanatory content or user benefit. The source-handover verification is refreshed after each archive change and recorded separately; it excludes external scientific/UI packages. Fresh science and original sentiment-candidate results are now recorded. Live student/LLM inference, final visual/institutional review and remote CI remain pending.

3.8 First-Mid Boundary

The supplied rubric evaluates the planned milestone, a meaningful live core module, requirements/design, testing, process evidence, report and student ownership. It sets no fixed whole-project implementation percentage. The preserved proposal places the scored prompted baseline and filtered training set at Mid-2 (week 18), and QLoRA with the three-way comparison at Final-2 (week 24). The first-mid commitments remain distinct from cached-record replay. Fresh ingestion-to-baseline/leakage execution and the original social comparison now provide actual evidence; unresolved scope, selection and review issues still need explicit closure.

3.9 Delays, Deviations and Recovery Plan

Repository access, original September bytes and exact-lock restoration still constrain historical reproduction; the new separately identified experiments execute. The portable artifact path permits tested explanation work while those inputs are restored. The assessment date is unconfirmed. The full mid package is estimated at 45–75 person-hours, subject to data and human-study access.

Chapter 4

Software Requirements

4.1 User Interactions / Use Cases

The increment lets a reviewer select a historical corpus/session, inspect all technical/news evidence, check explanations, inspect rejection/fallback, complete an explicit prototype risk profile, and compare a separate profile-dependent linear blend. It identifies absent social features, historical data, unchanged classifier probabilities and constant-Buy behaviour. The final retail-user journey remains broader than this demonstration.

4.2 Use Case Diagram

The implemented interactions are Select Historical Decision, Inspect Evidence, Check Explanation, Inspect Rejection/Fallback, Calculate Prototype Profile, Compare Separate Blend, and Inspect Social Study Evidence. The diagram represents the current reviewer/operator journey, not a complete approved retail-product design.

4.3 Detailed Use Cases

For Check Explanation, the precondition is an available, validated record. The user submits visible text and optional JSON claims. Both are checked against the record. A contradiction, missing essential display claim, malformed payload, or unsupported sentence in strict mode causes rejection and a verified deterministic fallback. An unavailable record produces an explicit error instead of a substituted decision.

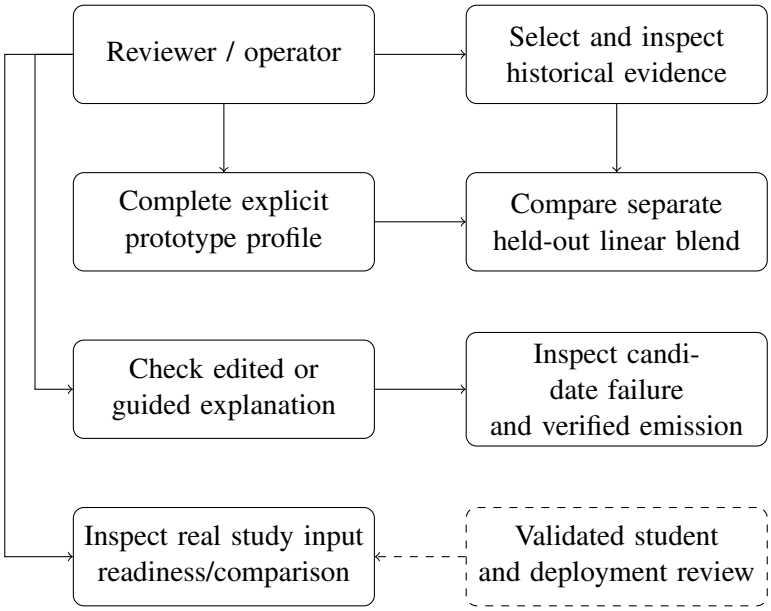


Figure 4.1: Implemented mid-evaluation reviewer/operator interactions and explicit study dependencies.

4.4 Functional Requirements

These identifiers document this increment; they do not replace the full proposal baseline or imply supervisor approval of a scope change.

Identifier	Requirement	Status
MID-FR-01	Load real saved decisions; disclose unavailable/invalid inputs.	Implemented
MID-FR-02	Show all technical/news features with values, attributions, meanings, and limitations.	Implemented
MID-FR-03	Check visible prose and JSON independently against the record.	Implemented
MID-FR-04	Reject known contradictions and missing essential claims; verify the fallback.	Implemented
MID-FR-05	Retain the candidate's rejection report separately from the fallback report.	Implemented
MID-FR-06	Reject unparsed sentences in strict mode.	Implemented
MID-FR-07	Disable scientific re-runs when input files are absent, empty, unreadable or structurally invalid, or runtime modules are missing.	Structural preflight tested

MID-FR-08	Request only an explicitly configured local generator; bound retries/timeouts and disclose fallback.	Adapter tested; real model pending
MID-FR-09	Check complete maintained feature definitions in both visible text and JSON.	Implemented
MID-FR-10	Preserve raw/failed attempts and frozen evaluation selection; report no model-quality score without completions.	Implemented; plan recorded
MID-FR-11	Check every feature's value, SHAP magnitude/direction/target, family totals and attribution scope in both guided representations.	Implemented
MID-FR-12	Require all profile answers; expose capacity/tolerance reconciliation and matched/binding project caps.	Implemented prototype
MID-FR-13	Fit comparator scaling on earlier development rows; refuse development-session decisions and missing technical inputs.	Implemented
MID-FR-14	Show the profile-dependent blend beside a fixed reference, preserving saved classifier probabilities and disclosing unvalidated assumptions.	Implemented prototype
MID-FR-15	Validate social-post provenance, declared AI or human initial labels and annotation provenance, adjudication/exclusions and a frozen chronological study plan.	Frozen AI reference; paired runs recorded
MID-FR-16	Require matching code/data/checkpoint/preprocessing and complete paired held-out outputs for sentiment comparison; preserve failures/raw runs.	Actual paired study; review pending
MID-FR-17	Run a declared social classifier only from existing local weights with explicit class/text/token mapping; retain logits, truncation and failures.	Both original classifiers executed
MID-FR-18	Retain actual teacher prose and provenance; verify it independently of assembled JSON; exclude test dates and export training/validation separately.	Six-example construction pilot; student pending

4.5 Non-Functional Requirements

The original quantitative NFRs remain separate from this increment's scoped engineering contracts. None of these planned performance bounds is accepted from template, fixture, HTTP health or import tests alone.

ID	Preserved original acceptance target	Current status
NFR-1	Feature assembly, inference, TreeSHAP and verification below 800 ms at p95; target VM, warm cache; excludes generation.	Unmeasured
NFR-2	Sentiment throughput at least 150 documents/s on 16 GB GPU, fp16 and 128-token input.	Unmeasured
NFR-3	TreeSHAP below 50 ms on CPU.	Unmeasured
NFR-4	Peak resident inference memory below 4 GB on the target VM.	Unmeasured
NFR-5	QLoRA within four GPU hours on the 16 GB envelope; later milestone.	Unexecuted
NFR-6	Explanation including verification below five seconds; target VM, CPU.	Unmeasured; reconcile
NFR-7	Nightly incremental ingest and scoring over the full universe below 30 minutes; GPU and local store.	Unexecuted
NFR-8	Full purged walk-forward sweep and hyperparameter search below 90 minutes on CPU.	Unmeasured

The local adapter's 20-second default socket timeout, configurable upper bound and two attempts do not establish a five-second whole-request deadline. Its bounded-failure behaviour is tested separately; NFR-6 is pending reconciliation and measurement. The inspected GPU has 10,240 MiB, so the proposal's 16 GB targets are not assumed satisfied. The separate recovered classifier runtime is CPU-only and is not a QLoRA environment.

Scoped ID	Testable increment acceptance	Evidence/status
MID-NFR-01	Preserve rejected-candidate reporting and verify fallback independently.	Verifier, emission and HTTP regressions
MID-NFR-02	Show explicit unavailable-data/refusal states without fabricated decisions or experiment results.	Record, readiness and widget tests

MID-NFR-03	Local generator requests use loopback, bounded attempts/socket timeouts and retain failures.	Transport/adapter fixtures; whole deadline unmeasured
MID-NFR-04	Expose only declared local routes, reject malformed/nonfinite/oversized requests and redact credentials.	HTTP/transport tests; prototype scope
MID-NFR-05	Git-free delivery verifies included inventory and reports uncommitted identity explicitly.	Source/handover tests and archive receipts
MID-NFR-06	A visible verification report must match the current record, text, JSON-inclusion and strict setting; obsolete record, source, profile and result requests cannot publish or release a newer request.	Eight pure-state assertions and twelve controlled Node handler cases; browser QA pending

These contracts do not replace the original NFRs or certify participant comprehension, deployment security or scientific accuracy.

4.5.1 Reliability

Missing inputs do not produce invented decisions. A template failing its own verifier raises an error. The portable server calls the real pipeline rather than simulating generation.

4.5.2 Usability

Candidate and emitted text are labelled separately. Feature meanings, raw-score attribution units, and unsupported capabilities are disclosed. Visual QA and participant comprehension checks remain pending.

4.5.3 Performance

Interactions perform no training or corpus scoring. HTTP duration is recorded in the live evidence JSON; it is not a browser-rendering or full-system latency measurement.

4.5.4 Security

The portable server binds to loopback and serves a single page, one explicit state script and declared APIs. It does not expose arbitrary project files or provide brokerage access, authentication, or production deployment.

4.6 Acceptance Criteria

Both corpora retain their recorded facts and accepted templates. Correct JSON cannot hide wrong visible prose; correct prose cannot hide wrong JSON. Rejection yields the verified template and retained failed report. Missing inputs are disclosed without synthetic replacement. Strict mode rejects unsupported sentences while preserving correct templates.

4.7 Requirements Traceability

The appendix and the machine-checked working traceability matrix map these 18 identifiers to exact source symbols and named executed tests. The check establishes anchor existence and execution, not a coverage percentage or institutional acceptance. Full proposal requirements still need reconciliation with agreed owners, approved iteration dates, social-study evidence, risk semantics, LLM evaluation, and deployment.

4.8 Mid-1 Priorities and Acceptance Matrix

The SRS remains a review draft. The following priorities select the implemented Mid-1 demonstration without silently removing broader proposal requirements. Must items define the live core module; Should items strengthen its explanation and evidence presentation; later items remain in the preserved proposal plan. Stakeholder/supervisor approval is not inferred.

ID	Priority	Observable acceptance	Source/check
MID-FR-01	Must	Unknown/invalid case displays no substituted class.	`load_cases` and invalid-record UI check
MID-FR-02	Must	Five technical/six news features remain inspectable with meanings and missing-value status.	`build_context`; both-family UI check

MID-FR-03/04	Must	Correct JSON cannot conceal visible 1.00% confidence; reject and check fallback.	verify/emit; injected-probability UI check
MID-FR-12/14	Should	No unanswered default; example profiles change comparator while classifier stays fixed.	`score_profile`; profile/reset UI check
MID-FR-15/17	Should	Both candidate scores use the same frozen 24 AI-reference posts and complete bound receipts.	`recorded_receipt_comparison`
MID-FR-18	Should	Actual teacher prose and four/two exports remain separate from student training.	`load_study`; teacher viewer
MID-NFR-06	Must	Editing text or changing case hides the previous accepted verdict.	identity-bound UI checks
UI-NFR-01	Must	Exactly one workspace and one walkthrough step executes per interaction.	current presenter UI test
UI-NFR-02	Should	No primary screen depends on remote fonts, corpus downloads or page-load model training.	offline styling and explicit action boundary

UI-NFR-01/02 are newly proposed presentation-quality acceptance conditions, not a change to the frozen scientific evaluation criteria. Full predictive/latency targets above stay unchanged. The observable checks demonstrate engineering behavior; human comprehension, projector usability and baseline approval require genuine review.

Chapter 5

Software Design Specification

5.1 Design Scope and Rationale

This design specification describes the implemented Mid-1 record, evidence, profile and explanation-checking module. The handbook permits the design models appropriate to the project; the selected component, domain, layered, data-flow, sequence, activity, state and deployment models answer different implementation questions. A relational database is not implemented: the schema diagram describes the actual JSON/Parquet artifact contracts and digest relationships. No screen or diagram implies a trained student, a production service or supervisor approval.

The key decision is to keep recorded classifier facts authoritative. A profile changes a separate retrospective comparator, and an explanation cannot overwrite model probabilities. The input validator checks probabilities, class argmax, feature/attribution consistency and missing values before a view appears. The verifier checks displayed text and structured claims independently; the emission path retains a rejected candidate while checking its replacement.

5.2 Overall Pipeline and Conceptual Domain

The fresh scientific increment establishes a real input-to-model path, rather than only an interface replay. Price/news admission, exchange calendars, strict-prior joins, label-end purge, fixed leakage arms, model/calibration and native SHAP produce retained outputs. The current presentation consumes those outputs without fitting a classifier. Original September bytes and historical vendor availability are not implied.

An equity study has dated market observations, derived feature evidence and a model run. A recorded decision case connects probabilities and attributions to a particular ses-

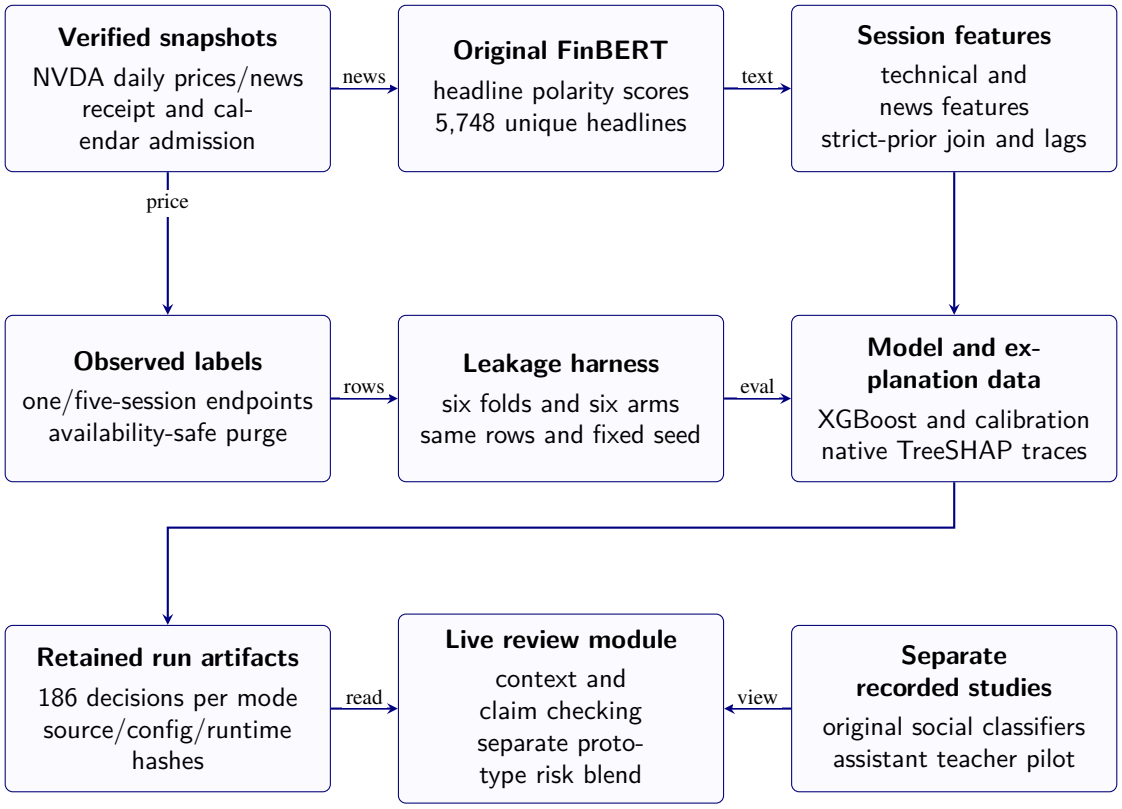


Figure 5.1: Current end-to-end boundary: separately executed price/news preparation, sentiment/features, observed-label evaluation and model traces feed recorded artifacts; the live module checks their explanations. Social/teacher studies remain separate.

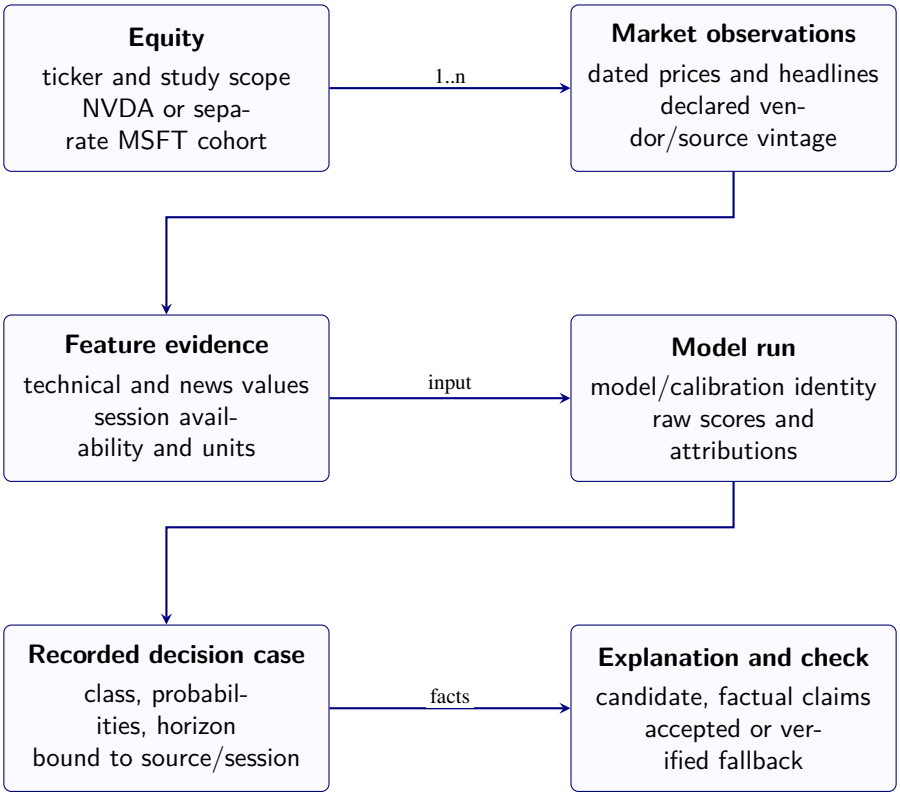


Figure 5.2: Conceptual equity-to-explanation domain. The current implementation realizes these concepts through typed records and file contracts; the following class model gives the concrete live-module objects.

sion/source. Explanation checking consumes that case. The prototype profile is an independent comparison, as the concrete class model below shows; no investor account or personalized classifier is invented.

5.3 Component and Layer Design

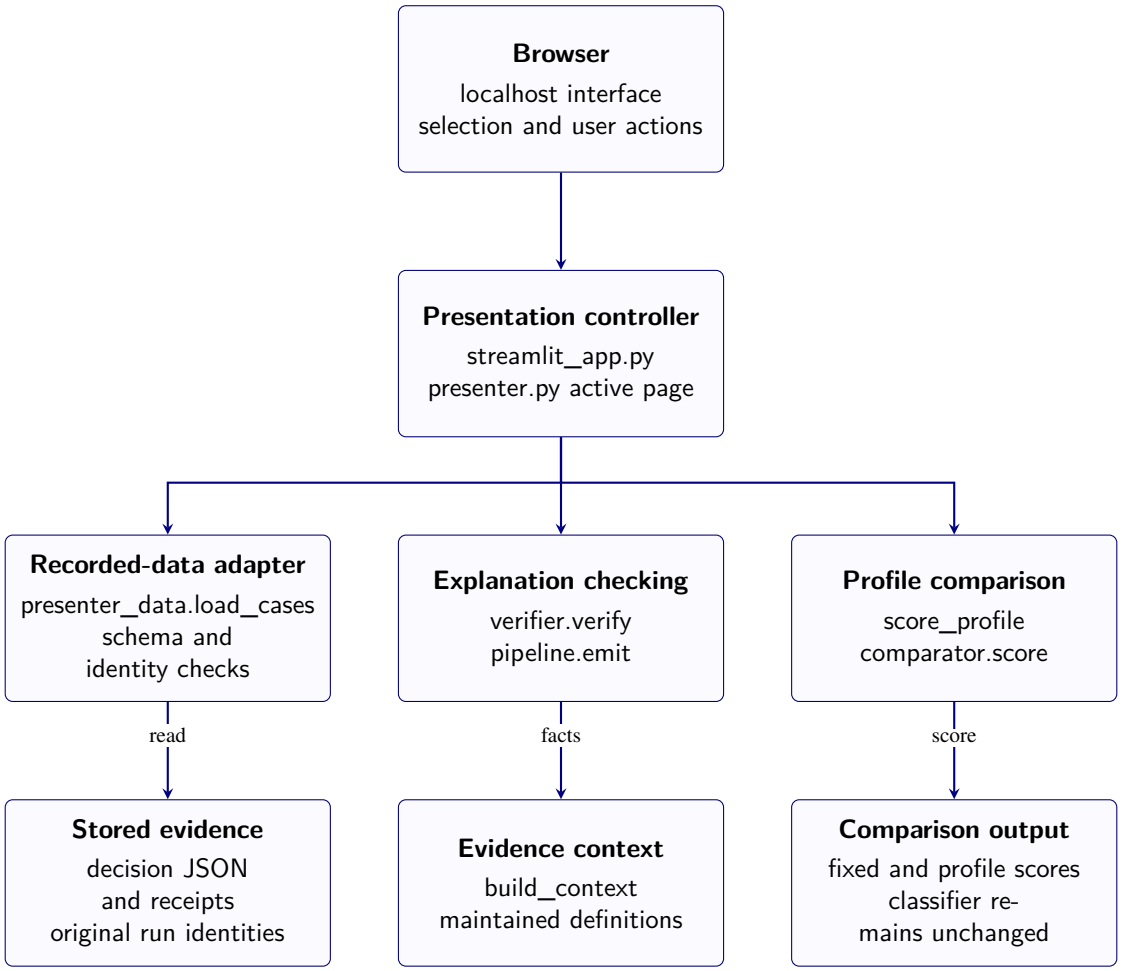


Figure 5.3: Implemented component dependencies. The presentation controller calls the recorded-data adapter, verification pipeline and independent profile comparator.

The main entry point is `app/streamlit_app.py`. `app/presenter.py` renders only the selected workspace and walkthrough step. `app/presenter_data.py` reads a fixed source registry, validates every recorded decision and refuses duplicate sessions, inconsistent model collections or changed bytes. A collection identity includes the full artifact hash; this prevents a check for one source revision from appearing as a check for another.

Streamlit provides the current presentation interface; the original standard-library HTTP demonstration remains an alternate historical interface. Neither opening the UI nor reading a study runs model training. Scientific execution uses its separately prepared runtime

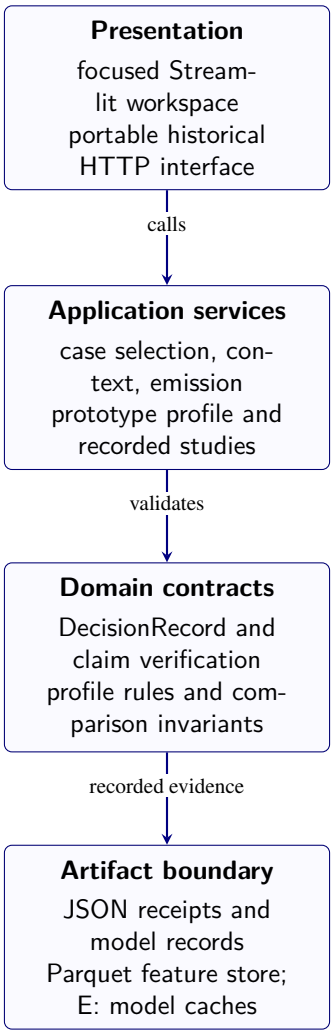


Figure 5.4: Responsibilities from presentation through application and domain contracts to recorded artifacts. Offline scientific execution is separate from page rendering.

and original verified publisher weights, while the current interface reads completed results.

5.4 Domain Model

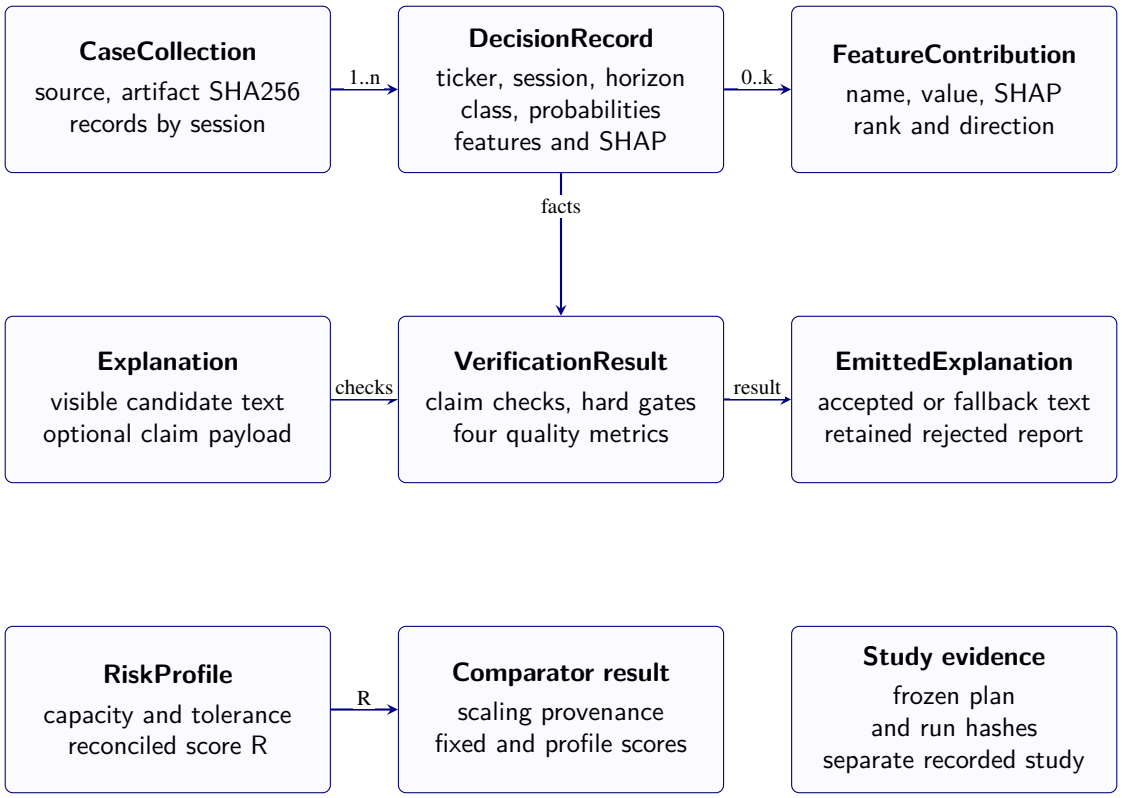


Figure 5.5: Implemented domain objects and their principal associations. A case collection owns session-indexed records; each record contains ranked feature contributions. Research-study evidence remains separate from the live classifier.

`DecisionRecord` is the factual boundary. Its dataclass rejects impossible probability or attribution states; nested dictionaries remain ordinary mappings and downstream functions treat them as read-only. `Explanation` contains visible text and an optional claim payload. `VerificationResult` retains per-claim verdicts and four metrics. `EmittedExplanation` carries accepted or verified fallback output plus the rejected candidate/report when present. A `RiskProfile` contains capacity, tolerance, caps and the reconciled score; its result is a separate comparison rather than a new prediction.

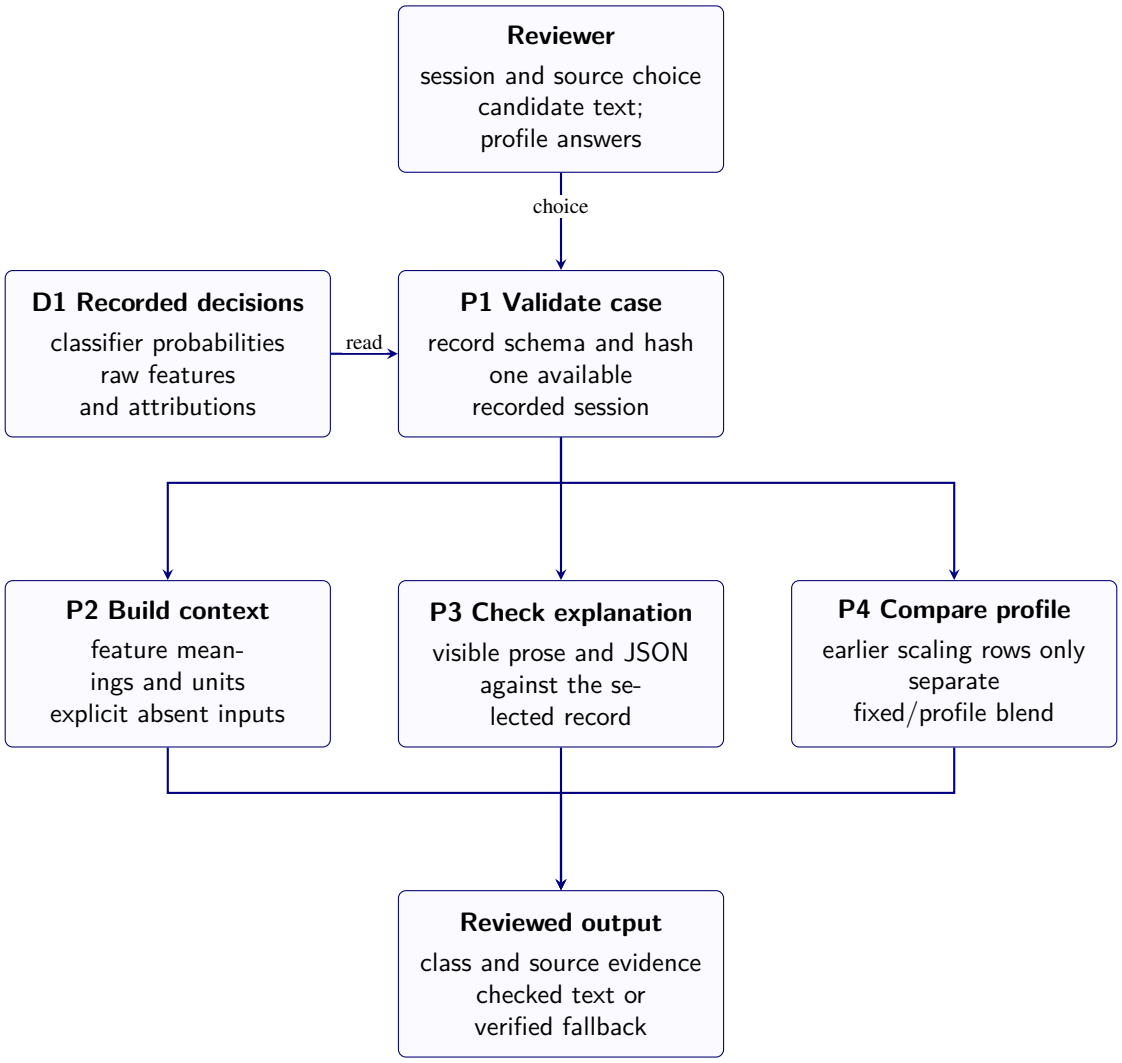


Figure 5.6: Logical data flow for a selected case. The current text and JSON meet at the verifier; profile answers flow to a separate comparator.

5.5 Data Flow and Artifact Schema

The reviewer supplies a source/session choice, optional profile answers and a candidate explanation. Validation reads the recorded data store, context supplies maintained meanings, verification checks the selected record, and the comparator uses earlier scaling rows. These paths return visible evidence, checked prose and comparison outputs. Social/teacher data is read only in its own research workspace.

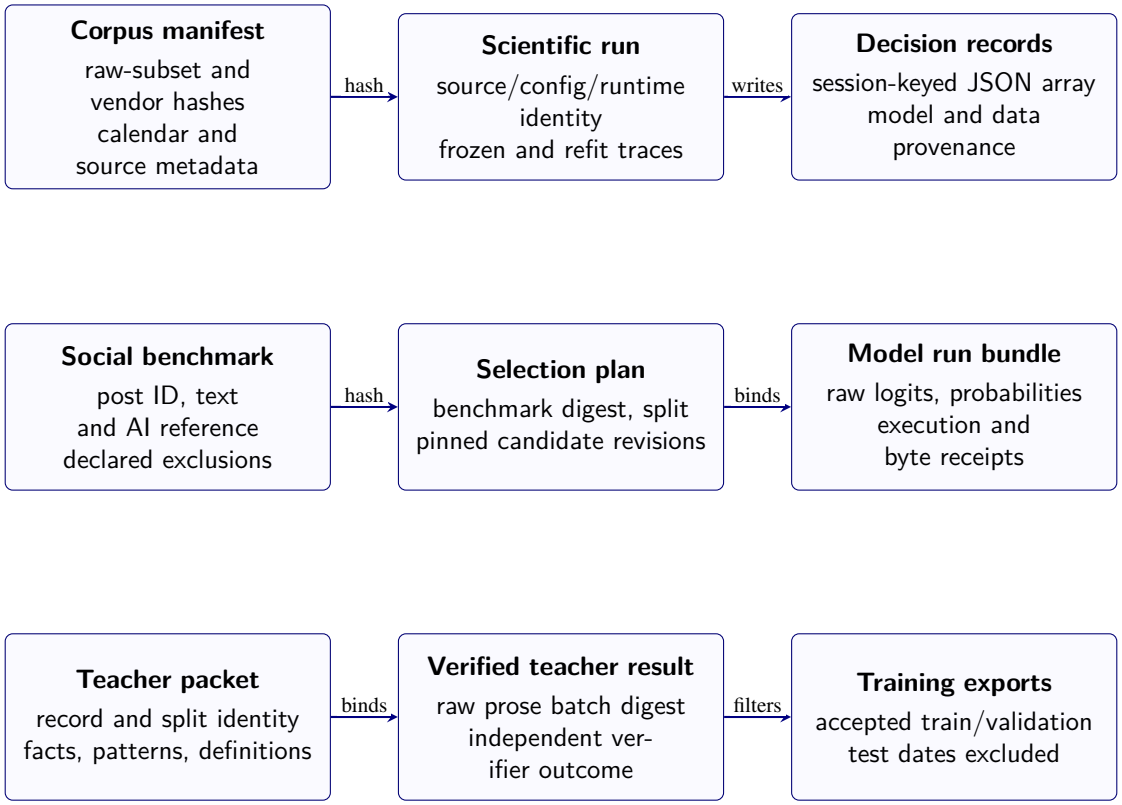


Figure 5.7: Actual artifact relationships expressed through source, plan, model and run digests. This is a file-schema design, not a claim of an implemented SQL database.

A scientific run binds input, configuration, source and runtime identities. Its model, scaler and calibration traces and decision records are retained under a new run directory. The social benchmark binds a frozen plan; each candidate bundle binds that plan, model revision, raw logits and execution receipts. Teacher packets bind record facts and split identities; accepted prose is exported to training/validation with held-out test dates excluded.

5.5.1 Data Dictionary

Item	Representation	Rule and meaning

Record identity	Source, session, artifact SHA256	One ticker/horizon/model per collection; sessions unique.
Class probabilities	Buy/Hold/Sell mapping	Finite nonnegative values sum to one within 10^{-6}; class equals argmax.
Feature vector	Named numeric/missing values	Five technical and six news features; absent evidence remains absent.
SHAP vector	Named raw-score contributions	Same feature keys; base plus contributions reconstruct raw output within 10^{-4}.
Top contributors	Ranked contribution objects	Nonzero contributions agree with feature values, magnitude order and direction.
Candidate explanation	Text and optional JSON	Visible prose and payload checked separately; known contradictions cannot be diluted.
Verification	Claim checks and metrics	Numeric fidelity, direction, coverage and unverifiable rate plus hard failures.
Profile	Five capacity and three tolerance answers	No unanswered default; caps and smaller-axis reconciliation disclosed.
Comparator	Scaling manifest and two scores	Earlier 80% rows fit scaling; later rows scored; no outcome accuracy implied.
Study reference	Post ID, text, label/exclusion	AI provenance declared; no independent human-label claim.
Publisher/run identity	Revision, SHA256, logits, receipts	Missing/changed weights, partial runs and incompatible metadata refuse comparison.

5.6 Behavior and Sequence Design

The controller receives the current candidate, not an earlier cached result. `explain/pipeline.py:emit` invokes `explain/verifier.py:verify`. A passing candidate is emitted; a failure keeps its report, renders the deterministic template and verifies it independently. If the template check itself fails, the operation raises an error and the UI claims no verified output.

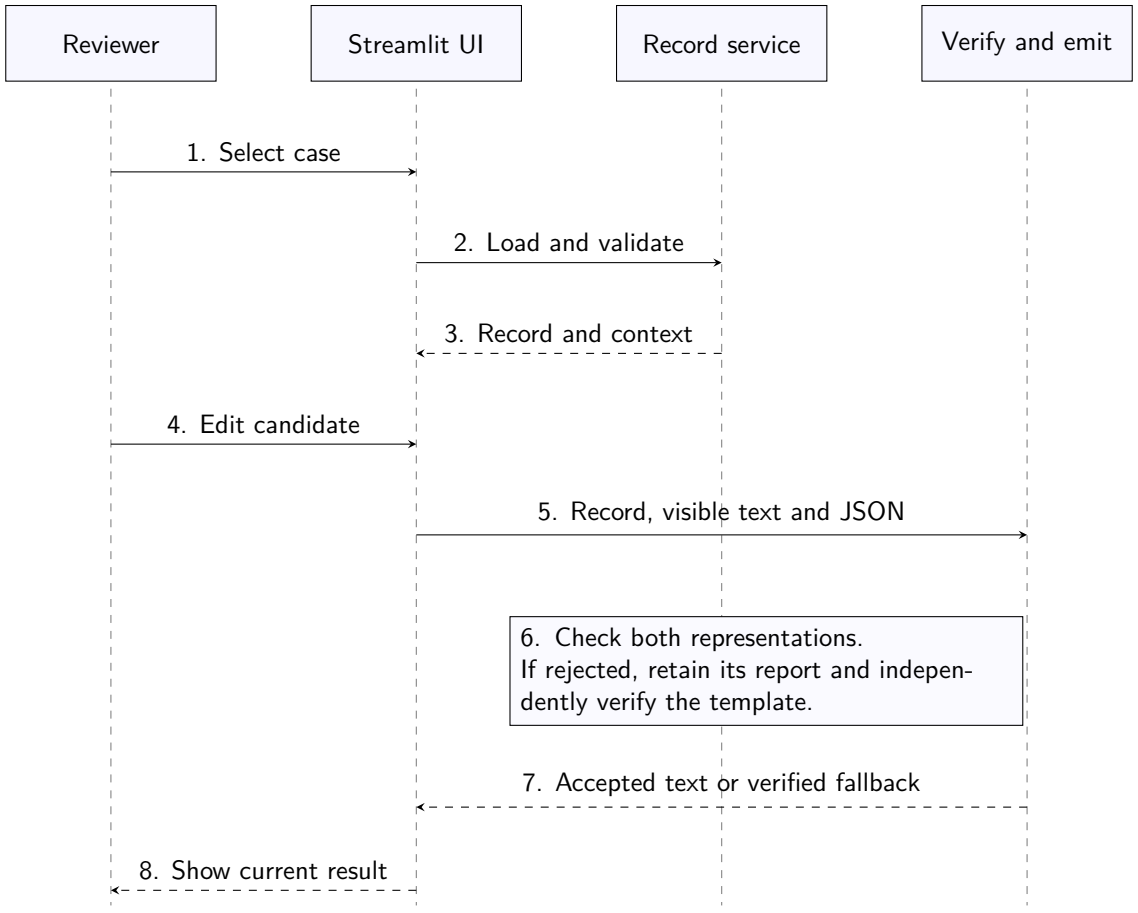


Figure 5.8: System sequence for the live explanation check, including the retained rejection and independently checked fallback.

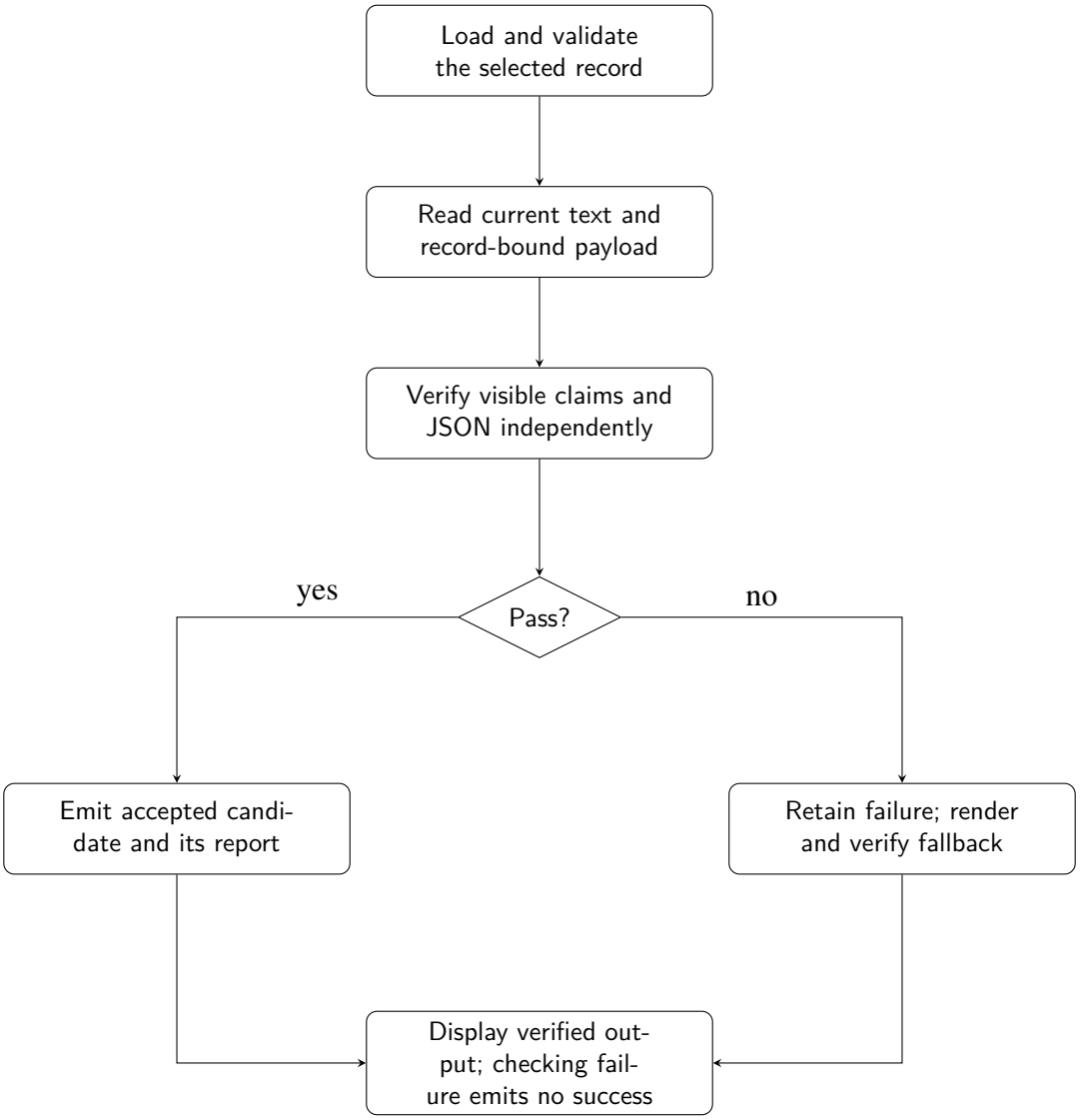


Figure 5.9: Activity flow and the two visible success paths. The failure path does not silently accept the candidate.

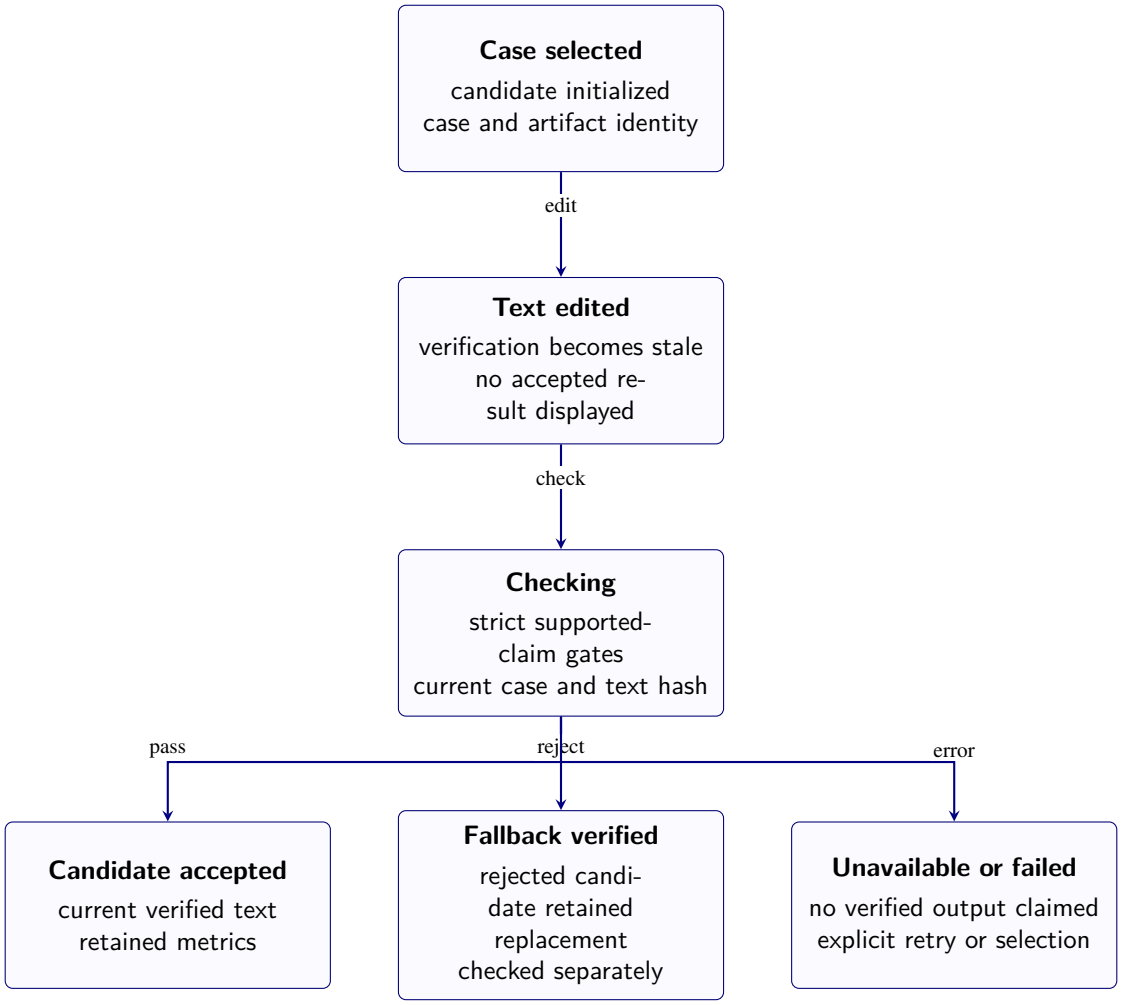


Figure 5.10: Explanation-view state machine. A case or text edit invalidates the visible check; only a result for the current identity can appear as accepted.

The candidate cache identity combines source, session, artifact digest and text digest. Changing cases resets the candidate to that record's template; editing text hides a previous verdict until checking runs again. Independent profile state can persist between steps without altering classifier facts.

5.7 Interface and Failure Contracts

Operation	Result	Failure behavior
Select recorded case	Validated record/context	Missing, duplicate, inconsistent or changed input shows an error; no substituted case.
Check explanation	Candidate/report or verified fallback	Known contradiction, malformed claims, missing essential claims and strict unsupported prose reject.
Apply profile	Reconciled profile and separate blend	Incomplete answers produce no score; development sessions refuse comparison.
Read study	Recomputed receipt-bound metrics	Invalid/partial evidence withholds comparison; no neutral substitute for failures.
Optional local generation	Explicit model request and checked output	Unconfigured model gives labelled deterministic guide; retry/timeout/identity limits remain explicit.

The portable historical API additionally exposes catalogue, record, verification, guide and profile/comparator endpoints. It rejects unknown routes, malformed/nonfinite JSON and oversized bodies. Those contracts and their live HTTP checks remain in the dated source handover; they do not establish browser visual quality or production security.

5.8 Deployment and Security Design

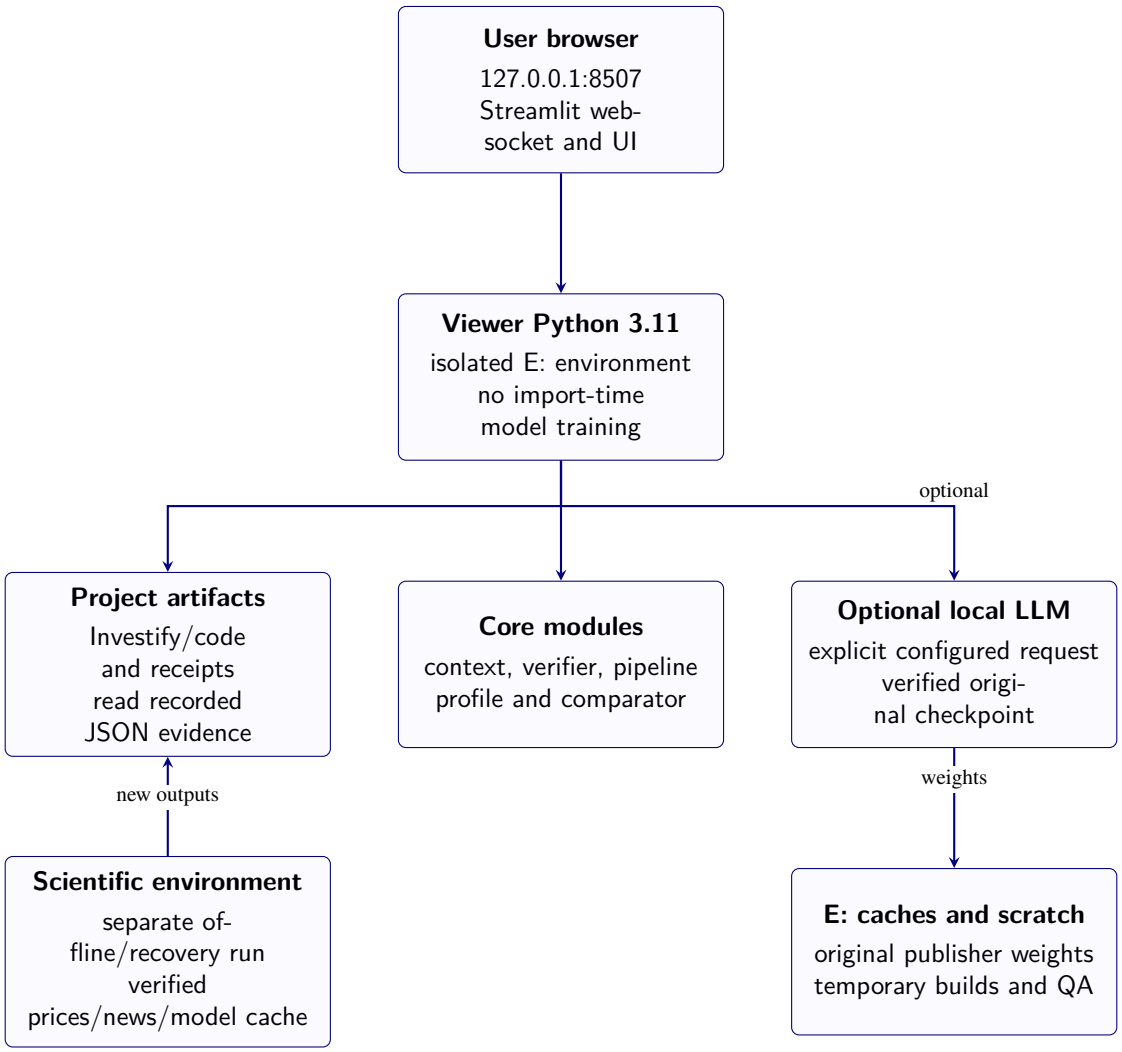


Figure 5.11: Current local topology. The viewer binds loopback; publisher weights and temporary builds stay on E:. Scientific execution and optional generation are distinct operations.

The launch script binds `127.0.0.1:8507`, disables usage telemetry and keeps cache/scratch on E:. It refuses an occupied port. The UI has no broker/account access, authentication or transaction path. Profiles stay in session memory; no profile database is implemented. Credentials stay in ignored local configuration and are excluded from review archives. This is a local review topology; a production target and authentication/deployment threat model remain later work.

5.9 Alternatives and Design Evolution

Decision	Alternative considered	Rationale and remaining limitation
One active workspace	All tabs mounted at once	Reduces competing controls and unnecessary hidden work; preserves explicit movement through the core demonstration.
Independent prose/JSON checks	Trust structured claims alone	A wrong displayed probability can coexist with correct JSON; the regression and live failure demonstrate the corrected boundary.
Deterministic fallback	Emit a failed candidate	Keeps supported record facts visible and retains the failure report; arbitrary language understanding is unproven.
Separate profile comparator	Relabel saved model output	Demonstrates prototype weighting without falsely claiming personalized XGBoost or suitable investor actions.
Artifact storage	Introduce SQL persistence	Actual interfaces consume immutable study files and transient profiles; adding a database would imply unsupported implementation.
Recorded studies in separate workspace	Train during presentation	Preserves reproducible source identities and a short core demonstration; live generation/training is not inferred.

The previous UI exposed four densely populated research tabs, then a guided wrapper. User review identified clutter; the present revision isolates the four-step case walkthrough and moves study details to their own workspace. This is an authorized presentation/design change. Scientific inputs, labels, model identities, thresholds and historical evidence remain unchanged.

Chapter 6

Iteration 1: Design and Implementation

6.1 Iteration Objectives

Preserve the existing project, validate the cached decision/explanation path, and make technical/news evidence inspectable. Establish the LLM input boundary and replay six recorded assistant-teacher examples; a live generator and measured user-comprehension benefit remain pending.

6.2 Selected Use Cases / Features

Select a historical corpus/session, inspect the decision/probabilities, read every feature's meaning and attribution, and inspect coverage/limitations. Fault checks reject corrupted confidence and show verified fallback.

6.3 Design

The frozen decision record remains the prediction source. A new builder combines its facts with reviewed definitions and implementation limitations. Streamlit and the standalone explorer reuse that context. Existing template verification remains the baseline.

6.4 Implementation

The active consolidated repository is on `main`; the earlier isolated branch is preserved in history. New code in `src/investify/explain/context.py` covers every feature, han-

dles missing values, rejects non-finite attributions/probabilities, and identifies unreviewed definitions. The Decision tab renders the context; the exporter uses the same records.

6.5 Completed Features / Module

The context, artifact tools, complete-guided verifier, report-retaining fallback, both live interfaces, prototype profile and separate retrospective risk comparator are implemented. Actual Streamlit widget tests pass. Six actual teacher explanations are saved and replayed in Teacher Study. Original social-candidate comparison now executes with recorded results in Chapter 8. Live LLM inference, social features in the saved predictor, personalized classifier predictions and remote GitHub execution remain pending.

6.6 Technical Implementation Details

Feature values and predicted-class SHAP contributions are separate. Family totals retain raw-score units. The verifier checks prose and JSON independently, including feature values and attribution targets. Displayed-text coverage, essential disclosures and hard contradiction gates prevent a correct payload from hiding an incorrect display. Strict mode rejects unsupported sentences. Evidence tools check schemas/context/templates/distributions/additivity and record artifact/runtime/source hashes.

6.7 Algorithms and Technical Decisions

Definitions follow actual features: dispersion is trailing variation of session-average polarity, and news attention is a headline-count z-score. Missing data stays missing. Indicator meaning and its use by this model are distinct [23, 24, 25].

6.8 Tools, Frameworks and Technologies

The portable path uses standard Python. The restored isolated viewer/test environment yields 212 selected passes and no skips; it is not the locked scientific environment. CI targets Python 3.11 on Windows/Linux with portable checks, frozen plans, comparator audits and a separate Streamlit AppTest job. Remote jobs have not run.

6.9 Demonstration Evidence

The continuation adds an explicitly labelled feature-level causality button: nine checks on synthetic fixtures detect four centred-window mutations, with no classifier fit. Scientific input/runtime absence disables the real re-run, and a later import/data failure is reported without a fabricated experiment result. Portable reports are invalidated after candidate/settings edits; record-loading errors retain selection/retry controls. These checks are distinct from browser visual QA. Artifact checks cover 186 records per corpus. The live HTTP exercise passes 62 explanation/risk cases. The separate comparator audit passes 228 arithmetic checks on actual held-out rows with labelled parameter scenarios. Chapter 7 records the current Streamlit browser review; these older portable checks have a separate scope. Both saved classifiers predict Buy throughout; no predictor-performance improvement is claimed.

6.10 Iteration Outcome

6.10.1 Versioned Scientific Corrections

The historical purger used calendar-day subtraction for observed-row labels. On the same labelled business-day fixture/test blocks it retains 18 training-label endpoints reaching the evaluation blocks; the corrected `observed-label-end-purge-v2` retains zero. Actual endpoints are calculated before feature filtering and carried to folds. Calibration candidates ending on/after first evaluation are excluded; a future-price shock leaves retained fixture inputs unchanged. Explicit frozen-classifier and refit-transfer modes now support a fair source-level comparison, with a genuine fresh Alpaca fitted comparison now recorded below.

Original scientific source/results are preserved. Cache joins refuse duplicate document keys or inconsistent probabilities, RSI edge windows have explicit behaviour, datetime units are normalized for the as-of join, and input ranges/ticker/revision/calendar contracts fail visibly. New outputs/cache directories are isolated. Full score/margin/truth arrays, calibration knots, scaler/model bytes and source/input identity are retained by the new trace mechanism. These earlier source/fixture repairs alone do not establish changed historical performance; the fresh-run findings below are separate. E_SAMEBAR_TEXT is a lag-rule ablation under the retained strict-prior join. The increment improves evidence coverage and transparency over the top-k numeric template and supplies a tested LLM input contract. It does not yet establish LLM usefulness, predictive skill, complete mid readiness, or deployment.

6.11 Recorded Assistant Teacher Construction Pilot

Under the user's latest direction, the current assistant supplies teacher examples before an alternative API is considered. The planned student remains Phi-3-mini; no additional Qwen teacher download is required. Six actual explanations were written from frozen historical NVDA records, and all six pass the current strict verifier. Input packets contain facts, reviewed definitions and controlled sentence patterns without a completed reference answer.

The supervision is hybrid: prose is actually assistant-generated, while the factual JSON payload is assembled deterministically from the original record. Both are checked, but the payload is not claimed as a teacher completion. Exact serving model ID/revision, decoding, token usage and per-example inference latency are unavailable and recorded as null.

Both corpora share date groups: 118 training dates, 30 validation dates and 38 preserved held-out test dates. The actual pilot has four training and two validation examples; test records are excluded. All saved source decisions are Buy, so neither class generalization nor adequate QLoRA data is established. Some held-out examples were previously explored; no first-ever untouched-test claim is made. Copied evidence replays original hash-bound raw persistence times and exact exports.

Original-student inference, QLoRA training, student quality and user understanding remain unmeasured. The inspected GPU has 10,240 MiB; the proposal's 16 GB envelope does not establish current-machine feasibility. Required training libraries, exact student checkpoint/tokenizer and measured token/VRAM budgets remain pending. Earlier Qwen-specific local pilot choices are inactive for the selected teacher/student workflow.

6.12 Fresh Alpaca Scientific Execution

On 12 October 2026 a separately identified, genuine Alpaca experiment executed the corrected pipeline. All 1,509 requested NYSE price sessions were present. The 6,768 revision-conservative news rows contain 5,748 distinct headlines, scored offline with original ProsusAI/finbert revision `0bc2ac10dda6929bcb1f77ccc43c7ec5c424f031`. The selected price basis is current-vintage SIP split adjustment, without cash dividend adjustment. News availability uses the later creation/update timestamp. Neither policy proves historical vendor availability. No FNSPID was downloaded; this run does not replace the preserved September results or reproduce their bytes.

Six arms at two horizons ran on shared row sets (1,309 one-day and 1,305 five-day rows), with six folds, fixed seed and unchanged model settings. The clean five-day macro-F1

is 0.2843 versus the training-distribution stratified-random baseline 0.3287, giving lift -0.0444 (95% fold-bootstrap interval [-0.0855,-0.0121]). The centred-Bollinger mutation gives five-day macro-F1 0.5390 and lift +0.2104 (interval [+0.1698,+0.2421]), demonstrating the deliberate canary. The one-day clean arm passes the configured control rule; the five-day clean arm remains below chance. E_SAMEBAR_TEXT remains a lag ablation.

Calibration mode	Eval. rows	Macro-F1	Brier before	Brier after
Frozen classifier	186	0.2238	0.7929	0.6105
Refit transfer	186	0.2238	0.7760	0.6078

Table 6.1: Fresh revision-conservative Alpaca experiment, five-day held-out evaluation.

Five of 186 candidate calibration labels ended on or after evaluation start and were excluded. The 181 retained rows fit isotonic mappings; 186 evaluation sessions span 29 March–21 December 2023. Both current modes produce 186 Buy decisions. Before calibration, the frozen model predicts Buy 40, Hold 51 and Sell 95. Its fitted isotonic mappings make Buy largest on every evaluation row. Raw held-out macro-F1 is 0.3182 in frozen mode and 0.3116 in refit mode; calibration reduces both to 0.2238 despite improving Brier. A fixed class-prior comparator estimated from the 181 available calibration labels has Brier 0.6095 on the same evaluation rows, versus frozen 0.6105 and refit 0.6078. This descriptive comparison does not establish a useful learned forecast; its probabilities use no evaluation labels. Thus classifier refitting is not necessary for the observed fresh-run collapse. This is an observation about the current experiment, not proof of the historical cause. Lower Brier and all-Buy classifications must be reported together.

Actual inference from retained model bytes/scalers exactly reproduces the stored raw scores. Independent native XGBoost TreeSHAP matches every exported contribution and bias; the maximum frozen additivity residual is about 1.02×10^{-6} against the unchanged 10^{-4} tolerance. Calibration, labels/endpoints and metric replay pass. Complete guided explanations verify on all 372 new records, separately from the historical 372.

The current source/input/runtime identities and execution logs are retained in `code/artifacts/scientific_runs/` and dated evidence receipts. The restored Python 3.11 runtime satisfies declared dependency ranges; it is not exact restoration of `uv.lock`. Engineering contracts record 179 passes and six legacy-input skips, with eight separate admission-refusal passes. No independently validated social accuracy, trading return, human benefit, student training or institutional approval is established.

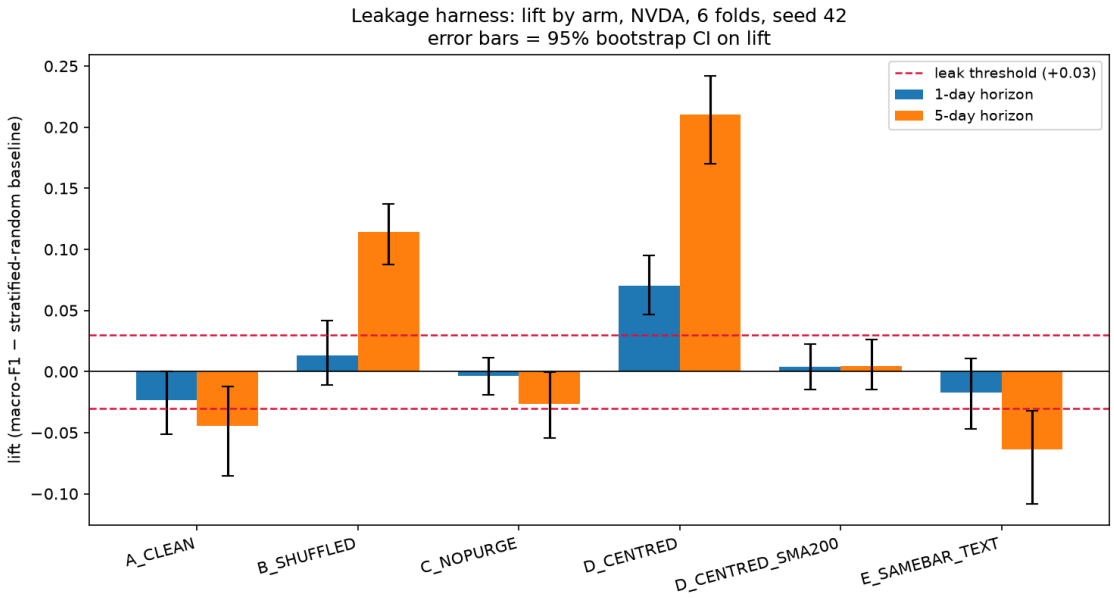


Figure 6.1: Actual fresh Alpaca leakage harness: lift relative to a training-distribution random baseline.

6.13 Data Structures, Algorithms and Code Conventions

The implementation keeps records indexed by session inside a source/digest-bound collection. Feature/probability/SHAP mappings retain named keys; ranked contribution objects preserve magnitude, value and direction. Context materializes JSON-safe feature rows without replacing absent inputs with neutral observations. Frozen plans and receipts use explicit SHA256 relationships rather than model-name assumptions.

The live checking algorithm is: validate the selected record; construct maintained context; read the current candidate and canonical facts; verify visible and structured claims independently; retain any rejected report; render and independently verify fallback; display only a result whose case/text identity still matches. The profile algorithm validates eight answers, computes weighted capacity/caps and tolerance, reconciles the smaller score, then scores only held-out comparator sessions using earlier scaling statistics. Neither algorithm updates the classifier.

New UI functions use snake-case names, explicit page/step registries and small functions named for user actions. Data-boundary docstrings state their no-training/no-write role. Syntax/AST checks and focused UI tests are engineering static/dynamic evidence; the group must still explain and modify its actual owned functions during review. The source/file walkthrough is supplied separately rather than replaced with programming listings in the report.

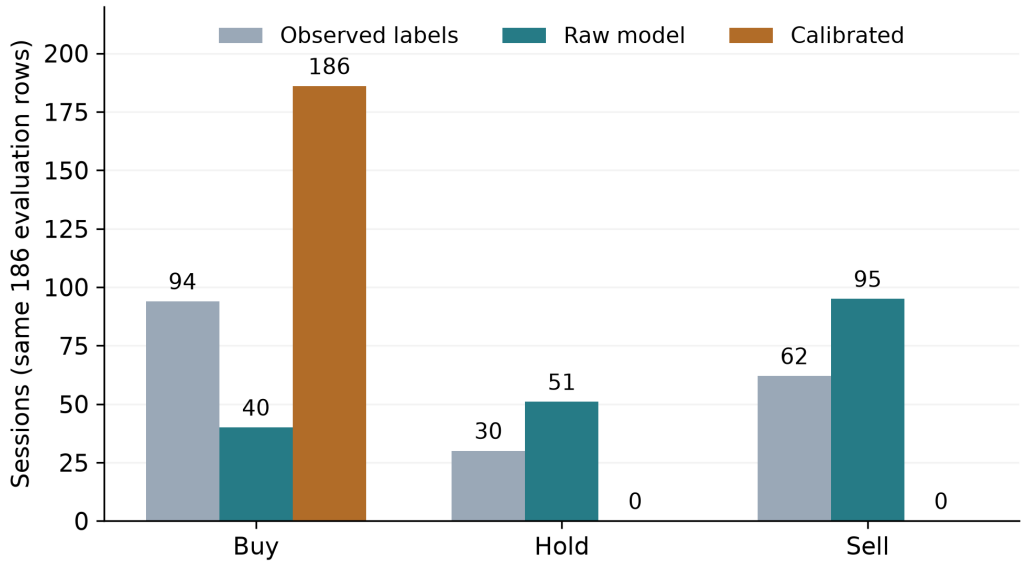


Figure 6.2: Observed labels, raw frozen-classifier predictions and calibrated classes on the same 186 evaluation sessions. Calibration collapses all emitted classes to Buy; this is descriptive retained-trace evidence, not a new selection or tuning experiment.

6.14 Mechanism of the Fresh Calibration Collapse

Both fresh modes share the same clipped isotonic maps. Buy's global output minimum is 0.5344828, while Hold and Sell never exceed 0.2500000 and 0.3139535. Buy therefore leads by at least 0.2205293 before normalization. Linear interpolation with clipped endpoints stays within these bounds. Dividing every output by the same positive sum preserves argmax: these maps force Buy for every valid raw probability vector.

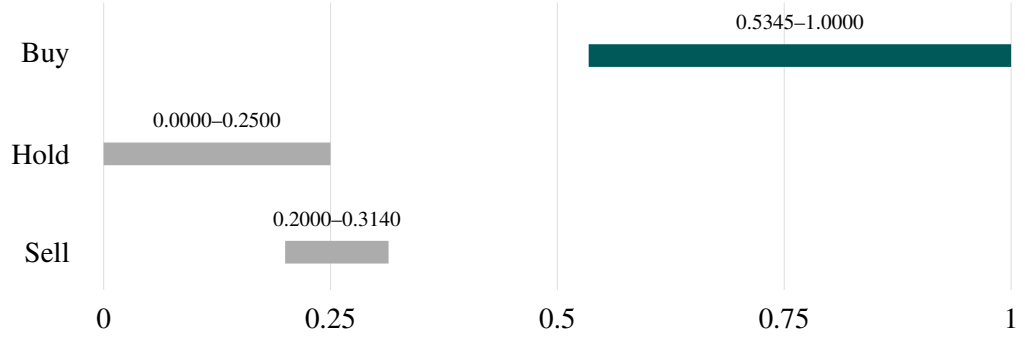


Figure 6.3: Global outputs of the fresh isotonic maps before normalization. Buy's minimum exceeds both competing maxima. The derivation concerns these byte-identified maps; the cause of preserved September outputs remains unproved.

Native replay reproduces 186 probability rows per mode and checks 5,151 simplex inputs per map set. Every output ranks Buy and sums to one; retained maps, labels and thresholds remain unchanged.

Chapter 7

Interface Design and Setup Manual

7.1 The Demonstration Journey

The redesigned interface has four workspaces: Overview, Walkthrough, Experiments and Design. Only the selected workspace executes. The core Walkthrough has four steps: Decision, Evidence, Risk profile and Explanation check. Technical tables, record provenance and optional model requests stay behind explicit detail controls. A persistent educational notice identifies the historical five-session research scope.

7.2 Core User Actions

Step	User action	Visible response
Decision	Select a recorded source/session.	Recorded class/probabilities and ranked raw-score contributions; all-Buy limitation visible.
Evidence	Choose technical or news family, then one feature.	Value, definition and contribution with absent social/freshness limits; all features remain inspectable.
Risk profile	Choose a labelled fictional example or complete eight answers.	Separate fixed/profile comparison; classifier remains unchanged; clearing removes profile output.

Explanation	Edit the candidate, inject wrong probability, then check.	Current candidate accepted, or rejected with independently checked fallback and retained report.
Experiments	Select leakage, original sentiment or teacher study.	Recorded outcomes and limitations; no page-load model inference.

7.3 Local Setup

The current project root is `E:/Uni_Stuff/FYP/Investify`. The viewer requires Python 3.11 and the separate demo dependencies, including Streamlit 1.63 or later. The restored viewer interpreter is on E:; it is separate from the scientific environment and does not imply restoration of the original scientific lock.

```
& '.\tools\start_guided_demo.ps1'
```

The launcher checks its interpreter and an available port, then binds `127.0.0.1:8507`. Supply `-Port` for another free port and `-PythonExe` for an explicitly prepared interpreter. It keeps cache/temp on E: and disables usage telemetry. A foreground launch stops with Ctrl+C; session hosting is not guaranteed after the process stops.

7.4 A Reproducible Failure Demonstration

Open the Walkthrough and keep fresh Alpaca frozen-classifier records with 21 December 2023. In Risk profile, the explicitly fictional cautious example has $R = 0$ and yields Hold in the separate comparator; the higher-tolerance example has $R = 1$ and yields Sell. The stored classifier remains Buy. The weighting is a prototype assumption, not a validated risk instrument.

In Explanation check, Inject wrong probability changes visible confidence to 1.00% while the structured payload retains the correct facts. Check explanation executes the same `emit`/`verify` functions used by the core tests. The candidate fails; the UI displays the independently verified fallback and keeps its rejection report. Editing text or changing the case invalidates the visible verdict. Restore record facts returns the factual candidate. Unsupported arbitrary prose can fail the strict maintained-claim grammar.

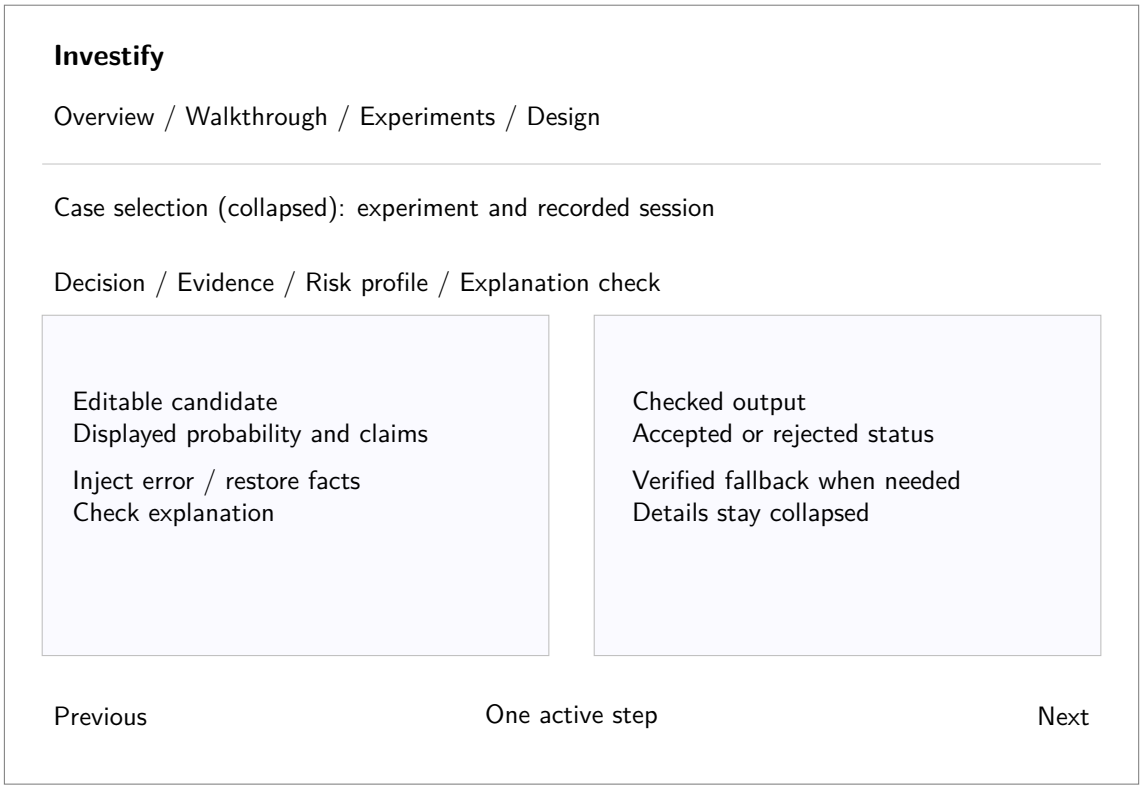


Figure 7.1: Design wireframe of the focused explanation-check step. This is a wireframe, not a browser screenshot; the report separately identifies actual captures when available.

7.5 Recovery and Boundaries

Missing or invalid recorded data yields an explicit error rather than a different synthetic decision. A development-block session refuses the profile comparator. Incomplete questions produce no assumed profile. Invalid study evidence withholds comparison scores. A failed checker claims no verified output. The optional local-model control makes a request only when explicitly invoked; an unconfigured model returns a labelled deterministic guide. Broker orders, accounts, profile persistence and trained-student inference are absent.

The portable historical interface remains a separately scoped alternative. Its old source/extraction/HTTP receipts certify their own snapshot; they do not prove visual quality of the rebuilt Streamlit app. The current headless Edge journey checks the implemented UI and retains original captures. Physical projector review and genuine rehearsal remain open.

An earlier browser capture exposed navigation beneath Streamlit's fixed header. The correction restores 4.75 rem clearance and full-width navigation. Fourteen current primary widget checks pass, including the optional source-bound calibration diagnosis. Their behavioral scope is separate from actual browser geometry and projector review.

7.6 Actual Browser Captures

The following figures are actual captures of the local Streamlit app on 13 October 2026. The desktop journey opens every walkthrough step, compares both fictional profiles, rejects the injected 1.00% confidence, and accepts the restored factual candidate. It also opens the recorded leakage, social and teacher studies and the Design view. The captured software layouts clear the fixed header and have no horizontal document overflow at the tested desktop widths. Tall capture viewports expose the complete content panel; they do not establish physical projector legibility or participant usability. Original-resolution PNGs and their hashes accompany the review bundle.

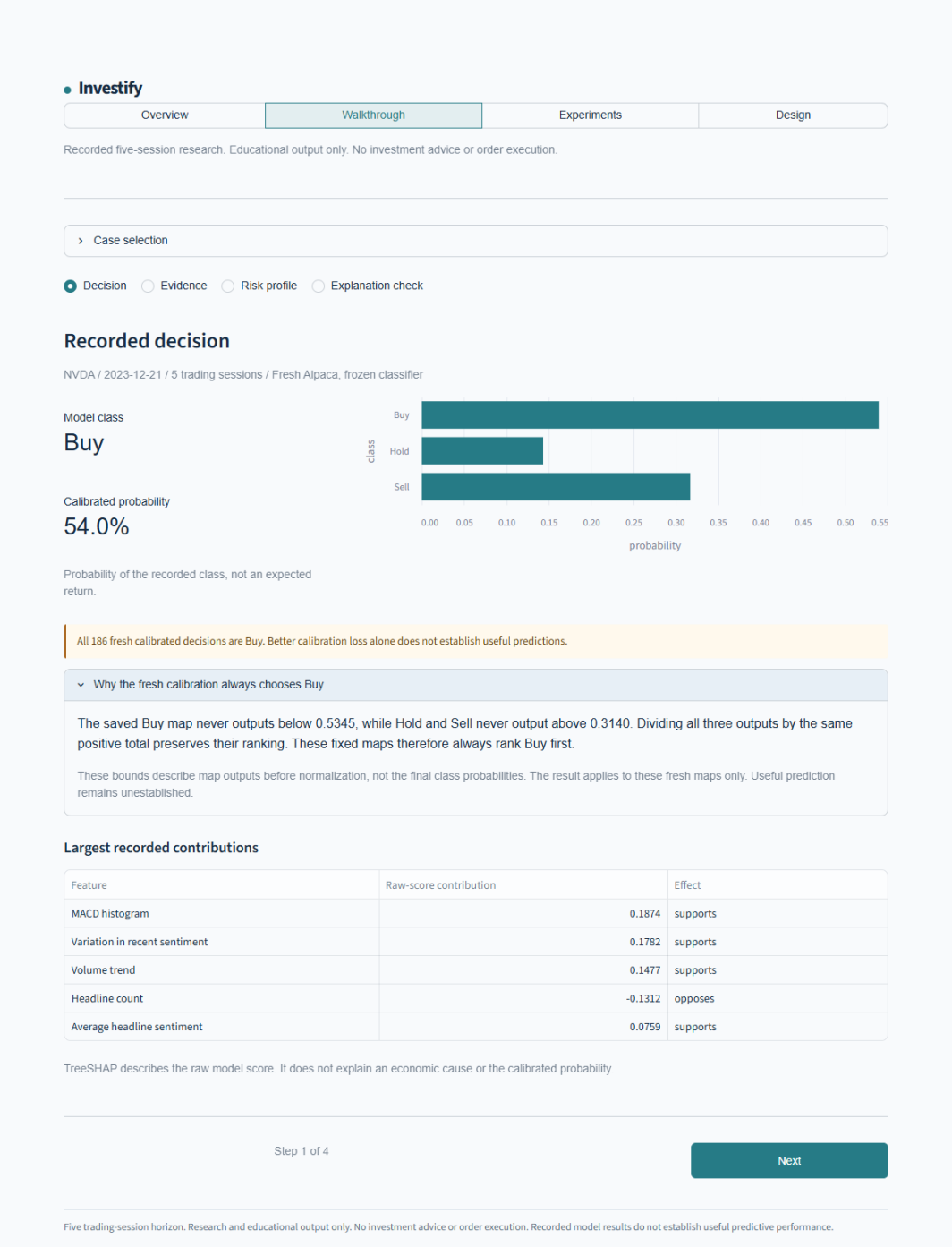


Figure 7.2: Actual decision screen with the fresh frozen-classifier diagnosis expanded. The displayed decision is Buy at 54.0% rounded confidence. The Buy map's pre-normalization minimum exceeds the other maps' maxima, explaining the constant class without establishing useful prediction.

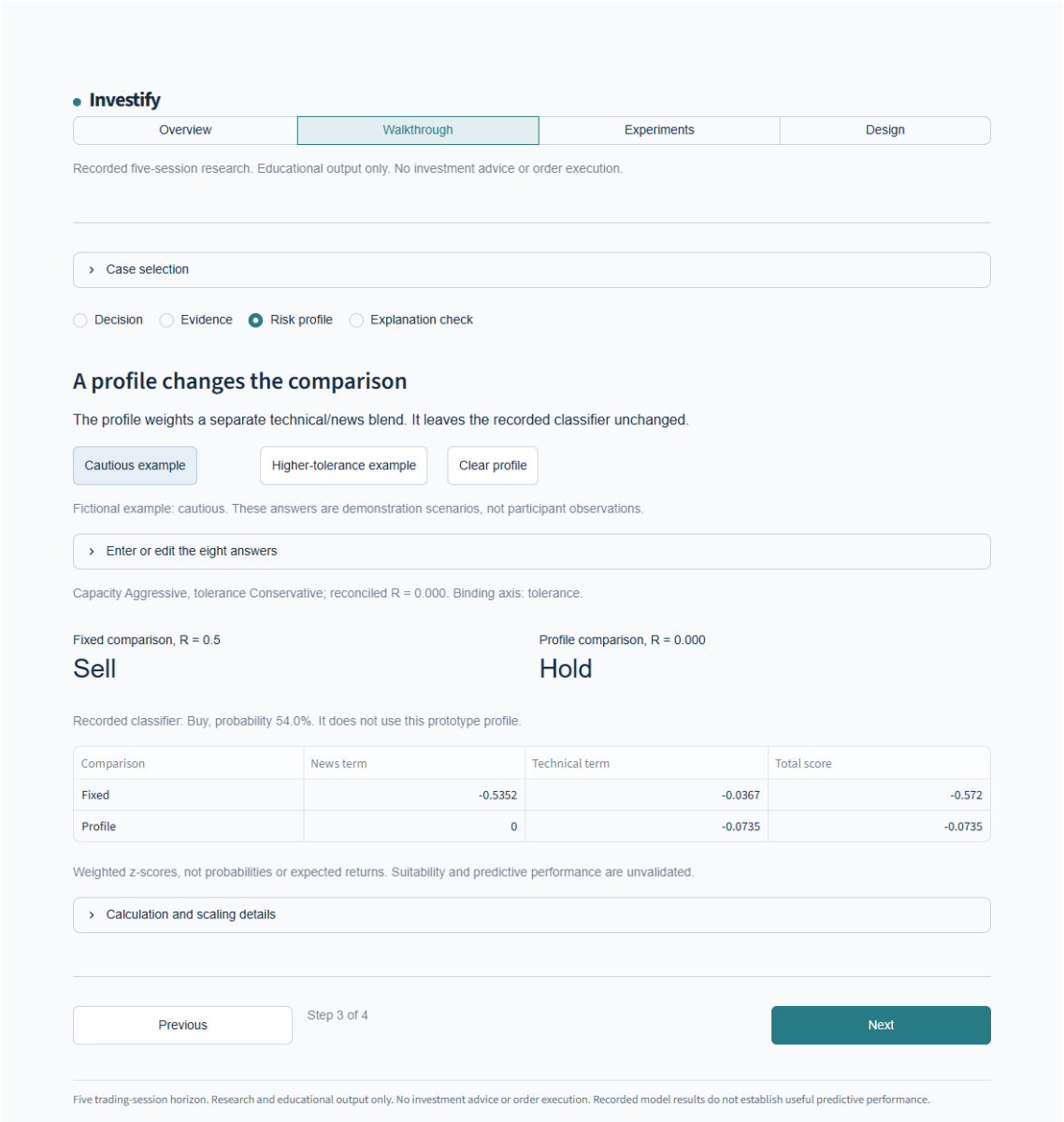


Figure 7.3: Actual cautious fictional profile: $R = 0$ produces Hold in the separate comparator while the fixed $R = 0.5$ comparison is Sell. The higher-tolerance $R = 1$ example produces Sell. The recorded classifier remains Buy at 54.0%; these prototype comparisons are not validated suitability results.

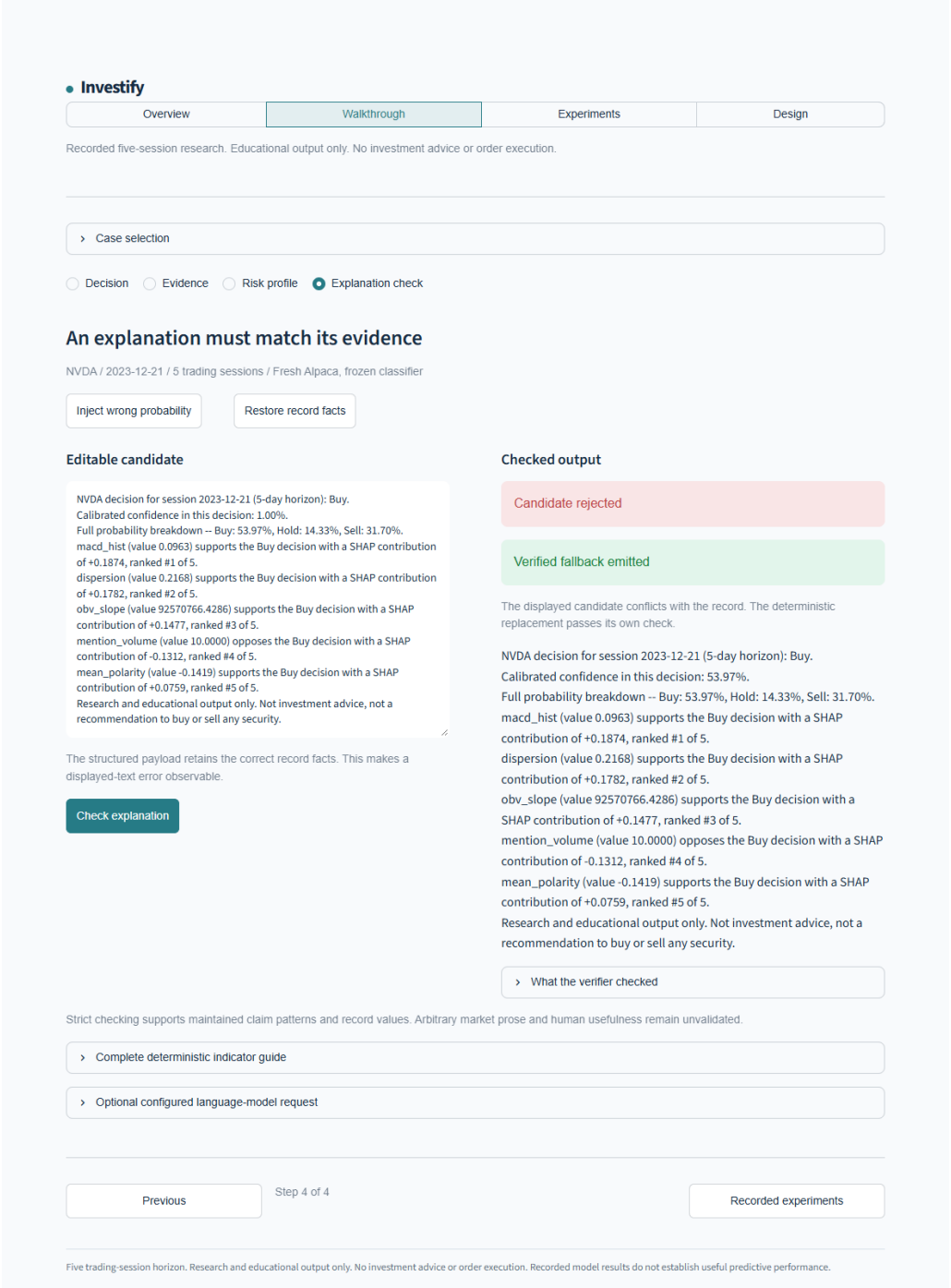


Figure 7.4: Actual explanation-check failure: visible confidence is deliberately changed to 1.00% while the JSON retains the record facts. The candidate is rejected and the independently checked fallback restores 53.97%. Restoring the facts and checking again produces an accepted candidate.

Chapter 8

Testing and Early Validation

8.1 Testing Strategy

Positive acceptance and labelled corruptions are both tested. Actual saved decisions support integration checks. Missing-input and corruption fixtures are test inputs, not synthetic substitutes for historical news or real LLM generations.

8.2 Test Levels and Techniques

Unit checks cover claim/schema validation, strict mode and fallback. Integration checks cover all 372 records. Local HTTP checks exercise the actual server and verification endpoint. Separate Streamlit AppTest cases exercise real widgets when its runtime is installed [26].

8.3 Test Cases

Important cases include correct JSON beside wrong displayed confidence, direction, value or class, correct prose beside wrong JSON, omitted essential claims, malformed/non-finite payloads, padding around a contradiction, unsupported sentences, and missing data/runtime. The appendix maps groups to requirements.

8.4 Test Results

Separate continuation core contracts have 153 passes with three shuffled-KFold cases excluded in the Python 3.11 viewer runtime. They use explicitly labelled fixtures, including

actual temporary Parquet writes and mocked model seams; they are not a full scientific or original-data suite. Six real-corpus loader cases remain skipped because their CSVs are absent. New V4 byte/receipt refusal and CPU tensor-serialization tests retain their own dated scopes; those earlier fixture receipts did not execute original weights; the genuine comparison below is separate. On 12 October 2026, the selected suite completed with **212 passed and zero skipped**, including 20 actual Streamlit AppTest cases. All 372 records retain schema/context/numeric-template and complete-guided-template acceptance. The live HTTP run passed 62 checks, including explanation and separate risk/refusal probes. The comparator audit passed 228 arithmetic/chronology/immutability checks on 76 actual held-out rows and three labelled profile scenarios. Runtime/source hashes and duration are in evidence JSON. This is not the complete scientific suite, a participant study or predictor-performance evaluation.

The default-mode fixture study has 112 cases: 80 corruptions rejected, 16 clean cases accepted, 16 vague-sentence cases accepted, and zero clean cases rejected. Vague-sentence acceptance remains a rate-threshold limitation. Strict mode rejects unsupported sentences but restricts accepted prose to the supported grammar.

8.5 Acceptance Testing / Validation

Programmatic acceptance is demonstrated for the explanation and prototype-comparator paths, including actual Streamlit widgets and HTTP health. The current browser journey has captured layout evidence. Physical projector review, participant comprehension and remote GitHub CI remain pending. No participant observations or member authorship are assumed.

The Social Study view's complete/partial flows are also exercised with clearly labelled benchmark/run fixtures: complete data exposes metrics and downloadable evidence; failed outcomes retain coverage/errors and refuse a paired comparison. Those widget tests do not establish actual sentiment-model quality.

8.6 Edge and Failure Cases

Correct JSON cannot hide empty text, an omitted decision or missing notice. Invalid claim structure selects verified fallback. Missing/zero feature evidence is preserved; non-finite authoritative probabilities or attributions are rejected. Missing records return errors; absent training inputs disable re-runs. The local adapter's unavailable-provider, malformed/truncated output and retry/fallback paths are tested with labelled fixtures. Incomplete profiles, development-date comparator requests and missing technical inputs are

refused. Actual-model behaviour still requires real generations.

8.7 Defects and Corrective Actions

Additional regressions expose the calendar-day/session-label purge mismatch, calibration outcomes reaching evaluation, Pandas 3 join-key resolution skew, in-batch duplicate persistence, delimiter-hash ambiguity, start-only news-bound exclusion, implicit UTC assumptions, cache-key row expansion, RSI all-rising missing values and stale portable reports. Corrections preserve historical source and raw/frozen evidence. Current package and trace receipts must match their exact source snapshot; earlier passing XML is not proof for later edits. New regressions reproduced payload-over-prose acceptance, unchecked feature values/target class, payload-supplied display coverage, and dilution by repeated correct claims. Independent checking and hard gates correct these defects. The old test requiring correct JSON to override wrong prose was replaced with the corrected regression. Rejection reporting now retains the failed candidate's metrics instead of showing fallback success as the rejection reason.

8.8 Current Validation Status

Nine social-selection unit tests use explicitly generated fixtures. They check declared post/annotation provenance, exclusions, chronological freezing, code/data/model identity, initial-label agreement, full paired outputs and null metrics on failures. The actual AI reference pilot contains 200 real MSFT tweets: 117 usable labels (31 positive, 67 neutral, 19 negative) and 83 explicit exclusions. Its frozen split is 93 development/24 held out, with only two negative held-out references. One session pass is recorded with full prompt/input/raw-output/timing and declared model-identity provenance; agreement/kappa are null and independent human validation is absent. The later original-publisher paired comparison is recorded below. These tests are not sentiment-model study results, and schema validation cannot authenticate human independence.

Eight offline-inference tests use a labelled backend fixture. They check actual held-out text routing without human labels, declared preprocessing/class mapping, stable softmax, rejected incomplete/mismatched heads, unknown plans/models, missing weights and retained errors/truncation. Those early fixtures did not execute original weights; actual publisher-byte-bound Torch/Transformers inference now completes both candidates, as recorded below.

Python 3.11.9 and an isolated viewer environment were used, not the locked scientific environment. The viewer was restored from existing E: cached wheels plus two un-

changed vendor pure-Python dependency sources/fallbacks, with provenance/licenses retained. Streamlit 1.63.0, NumPy 2.4.6 and Pandas 3.0.6 are frozen separately; dependency consistency passes. The original September raw/store inputs remain absent; a separately identified fresh Alpaca experiment now executes on restored genuine inputs. Local model probes returned no available identifier/completion; no LLM generation is claimed by those probes. The separate fresh science and original social-classifier runs retain actual execution evidence. The report is compiled as an unsigned working draft with layout inspection recorded separately; institutional review is incomplete. Both saved classifier folds remain constant-Buy.

8.9 AI Reference Pilot and Saved Calibration Audit

The actual AI reference pilot contains 200 real MSFT tweets: 117 usable labels (31 positive, 67 neutral, 19 negative) and 83 explicit exclusions. Its frozen split is 93 development/24 held out, with only two negative held-out references. One session pass is recorded with full prompt/input/raw-output/timing and declared model-identity provenance; agreement/kappa are null and independent human validation is absent. The later original-publisher paired comparison is recorded below. Ten additional annotation/import unit tests use labelled fixtures to check raw packet coverage, input/prompt tampering, single-pass null agreement, optional cross-checks and explicit exclusions. The actual raw-label batches are separate research artifacts. The Social Study widget also checks the AI-reference disclosure and missing-inference state.

The saved-artifact audit independently summarizes the preserved 372 historical records, all Buy; their historical cause remains unproven. The later genuine revision-conservative Alpaca experiment executes both corrected calibration modes on matching evaluation dates. Each still produces 186 Buy decisions. Actual retained-model, isotonic, label-endpoint, metric and native-TreeSHAP replay passes, and every new guided explanation verifies. The raw frozen classifier's diverse decisions collapse after its fitted isotonic mappings; refit transfer is therefore not required for the fresh observation. Five unavailable calibration targets are excluded. The new experiment's clean five-day score is below its random baseline, while the deliberate centred feature canary fires. Chapter 6 records exact metrics, input/runtime limits and the distinction from September results.

The frozen-reference evaluability audit selects neutral as the development-majority comparator, obtaining 15/24 exact held-out AI-reference agreement and macro-F1 0.256410, with zero positive/negative recall. This deterministic comparator is not candidate inference or independently human-validated accuracy. One held-out prediction changes exact agreement by 4.17 percentage points; negative recall moves in 50-point steps. No pair meets the declared token-Jaccard 0.8/minimum five-token heuristic, which does not es-

tablish independence or exclude shared events/pretraining overlap. Reference and split hashes are unchanged. The separate generative-model replacement remains pending verified weights and available access; the original-classifier study below is complete.

8.10 Teacher Pilot Evidence

The six recorded examples, four-training/two-validation split, serving-identity limitations and untrained Phi-3-mini student are documented once in Section 6.11. No training efficacy or human-comprehension result is inferred.

8.11 Executed Original-Publisher Social Comparison

On 13 October 2026 both pinned original classifiers completed the same 24 chronologically held-out MSFT posts. The frozen reference contains 7 positive, 15 neutral and 2 negative AI labels. Source posts, reference labels, exclusions, candidate identities, preprocessing and the original V3 split are unchanged.

Candidate	Macro-F1	Exact agreement	Positive recall	Negative recall
CryptoBERT	0.2564	15/24	0/7	0/2
Cardiff	0.6016	16/24	4/7	1/2

Table 8.1: Agreement with frozen AI reference labels; not independently human-validated accuracy.

CryptoBERT predicts neutral on all 24 posts, matching the development-majority neutral comparator. Cardiff is the exploratory benchmark leader. A single prediction changes exact agreement by 4.17 percentage points, and negative recall changes in 50-point steps. Public-post pretraining overlap and shared events remain unexcluded. These results do not establish NVDA transfer, return prediction, significance, suitability or deployment readiness. No predeclared adequacy threshold exists in the retained plan, and the deployment selection remains unset pending actual criteria and review.

Both original publisher revisions, weight hashes and all loading-affecting tokenizer/config files are verified. Original Cardiff BIN weights were losslessly serialized: all 204 tensor keys, dtypes, shapes and raw bytes match. Its two unused native pooler parameters are retained during modern loading, with exact classifier-logit parity verified. No classifier head is missing, mismatched, substituted or randomly initialized. Modern set-valued loader diagnostics are normalized for recording; nonempty diagnostics still refuse inference.

The first Cardiff loading refusal is retained. Both models reran under a new source-bound plan, retaining per-post raw logits, processed-text hashes, token counts, truncation, run-

time identity and probability outputs. Independent receipt/logit comparison reproduces the reported table; repeated CryptoBERT outputs are identical. The Social Study view displays recomputed recorded scores with AI-reference limitations and download evidence. The earlier recorded-social scope has 25 viewer/selector passes, including 21 Streamlit cases; the later guided-demo scope has 28 passes, including 24 AppTest cases. These are overlapping suites; 37 current loader/pooler checks and nine subtests pass on their declared scopes. These scopes overlap earlier evidence and are not summed as a full-suite total.

8.12 Current Rebuilt Interface Validation

The new one-workspace interface has its own current AppTest scope, separate from the retired mounted-tab interface. Twelve tests exercise route isolation, both fresh and archived sources, both feature families, explicit/cleared profiles, development-session refusal, visible-probability corruption and fallback, stale text/case results, invalid records, actual recorded research scores and failed-study refusal. The first current run has twelve passes. Browser/projector inspection is separately recorded when performed; no participant acceptance is inferred.

8.12.1 Test Plan and Release Checks

Level	Technique	Planned/retained evidence
Unit/domain	Equivalence partitions and boundaries	Finite probabilities, class argmax, attribution consistency, malformed claims and profile combinations.
Module	Controlled fault injection	Correct JSON beside wrong visible confidence; missing essential claims, unsupported prose and fallback failures.
Integration	Real recorded inputs	Context/template checks on retained records; current case identities and unchanged classifier facts across profile actions.
System	Actual UI and local HTTP	Current AppTest interactions and separately dated portable server endpoint/transport checks.

Scientific	Chronological controlled comparisons	Fixed seed and row sets, six folds/two horizons, deliberate canary, model/scaler/calibration/native-SHAP replay.
Usability and accessibility	Browser/projector walk-through	Inspect screen density, readable labels and complete flow; genuine person/rehearsal evidence remains pending.
Acceptance	Supervisor/member walk-through	Confirm baseline, actual ownership and proposal-action closure; no automated test substitutes for these approvals.

Before delivering a revision: check source syntax, execute relevant changed-path tests, verify evidence/source hashes and secret exclusion, compile/render the report and inspect every changed page, inspect actual UI captures when available, and create a new source snapshot. Old receipts are not overwritten or summed into a full-suite number.

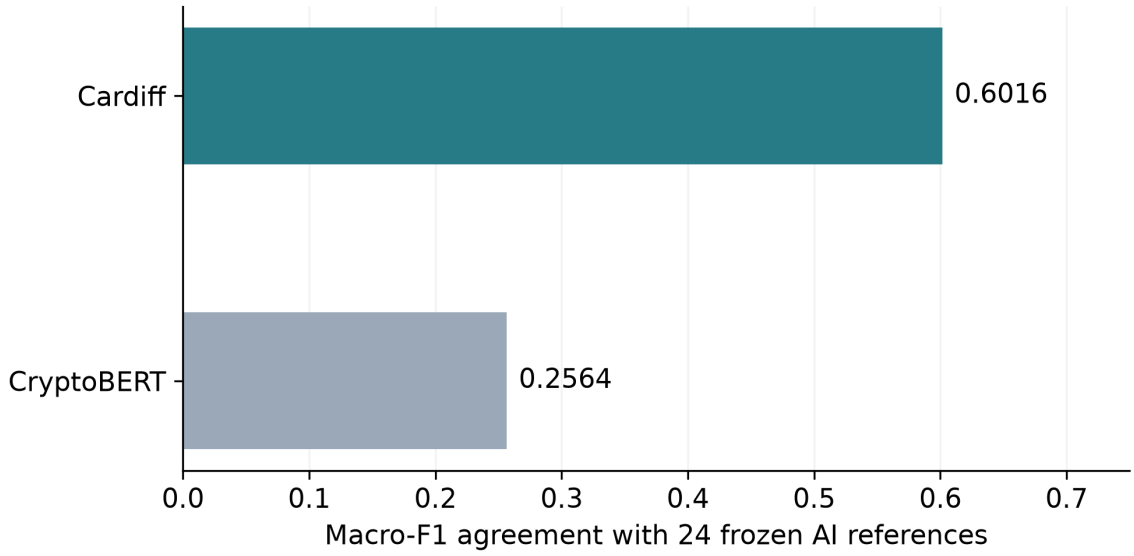


Figure 8.1: Original-candidate macro-F1 agreement against the same 24 frozen AI references. Only two negative references and absent independent human validation limit interpretation; deployment selection remains open.

8.12.2 Header and Calibration-Diagnosis Revision

The current scope has 42 passes: fourteen primary-interface cases, twenty-four preserved legacy cases and four recorded-social selectors. Added checks expose actual map bounds

and refuse stale trace identities. Earlier receipts retain their own sources and are not summed into this total.

Ruff 0.14.0 checks the current UI modules and tests, together with the calibration, browser, packaging and delivery tools. E4, E9 and F rules have zero unresolved findings. Two E402 annotations explain source-path bootstrap imports. The pinned checker is in prepared CI; remote execution and genuine member acceptance remain open.

The separate actual Edge journey captures the four-step walkthrough, two explicit profile examples, explanation rejection/fallback and restored-candidate acceptance, the recorded studies and Design. Desktop document widths and header clearance are checked; original PNGs are visually reviewed. This supplements AppTest without adding those interactions to the 42-test count. It does not establish physical projector legibility, participant usability, human understanding or supervisor acceptance.

Chapter 9

Project Management and Development Evidence

9.1 Work Completed

The isolated copy preserves original history/documents. Completed work includes complete feature/family evidence, independent prose/JSON checks, strict mode, rejection/fallback reporting, local LLM transport/evaluation plumbing, both live interfaces, restored viewer runtime, authoritative-record validation, an explicit prototype profile and a separate retrospective risk comparator. The selected suite has 212 passes and no skips; local evidence covers 372 records, 62 live HTTP cases and 228 comparator arithmetic probes. CI jobs are prepared.

9.2 Work Remaining

The real fresh Alpaca scientific run and original-publisher paired social comparison now complete, with retained raw execution, source/input identity and replay evidence. Cardiff leads the small AI-reference benchmark; final adequacy criteria, independent validation and deployment selection remain open. Source/raw labels, exclusions and the original V3 chronology are preserved.

Current gaps are institutional report/deck inspection and completion, genuine member contributions and review/rehearsal, current repository/remote CI evidence, live LLM usefulness and agreed risk semantics. The original September bytes and exact scientific lock are not reproduced. Student inference/training and broader acceptance work remain later milestones.

9.3 Contribution of Each Team Member

Section 3.6 contains the proposal's planned member assignments. Ahmed covers the point-in-time harness, leakage/timestamp/parity work and deployment; Rafay covers risk profiling, sentiment, degraded operation, acceptance and frontend; Soban covers the decision engine, calibration, SHAP, verifier and later QLoRA. Each member must record actual implementation, review, testing and defence preparation. The planned table does not establish completion or authorship of assistant-supported work.

9.4 Repository and Version Control Evidence

The consolidated repository is on `main`. Original baseline `1dc24e0` and its isolated branch are preserved in history; local commit `93d0d60` records the fresh scientific increment. Current verification evidence records source/data hashes. Upstream access remains unverified; local workflow configuration is not evidence of a successful remote CI run.

9.5 Continuous Integration and Delivery Status

The GitHub Actions configuration runs two Python 3.11 jobs on Windows and Ubuntu after a push, pull request or manual dispatch. Both call the same versioned `run_mid1_ci.py` entry point used for local replay. The portable job executes standard-library regressions, saved-record validation, preview generation, a plan-only explainer check, comparator arithmetic and a real localhost HTTP exercise. The UI job runs scoped source checks and the legacy, presenter and recorded-study selector suites. Each retains logs and a byte-bound source/artifact receipt; CI artifacts are configured for fourteen-day retention.

The Windows local replay passes both jobs: 165 portable tests and seven CI source-ownership/metadata guards, 62 live HTTP checks, 372 validated historical records and 228 comparator probes, plus 42 UI tests in separate overlapping scopes. Temporary component metadata is restored after each job. The replay uses the already prepared local runtime; it does not establish a fresh dependency installation or Ubuntu/GitHub execution. The consolidated repository still has no configured remote. Actual remote CI remains open, and no continuous deployment, hosting target, model training or paid inference is executed by this workflow.

9.6 Milestones and Commit Evidence

The checkpoint distinguishes the initial context increment from verifier/local-demo work. Historical experiments are preserved separately from current test/HTTP evidence. Reproduction instructions appear in the repository appendix.

9.7 Tools and Tool-Assisted Development

The coding assistant contributed code, regression tests, drafts and programmatic validation. Earlier handoffs declare GPT-6.1 Sol/Xhigh as selected session configuration; the exact serving revision is not provider-authenticated. Student review, source verification, institutional declarations and accurate tool disclosure remain necessary. Student authorship is not inferred from assistant work.

9.8 External Components and Dependencies

The baseline uses FinBERT, XGBoost, isotonic calibration, TreeSHAP and Streamlit. The portable demo uses standard Python HTTP/JSON facilities and existing record/verifier code. The retained social labels were generated by the assistant in one actual session pass and disclosed as AI references. Six recorded assistant-teacher explanation completions are retained separately. No API/local student inference or human annotation is claimed.

9.9 Risks and Mitigation

Both interfaces expose recorded evidence and failure cases. The restored scientific runtime executes a separate fresh snapshot; original-byte and exact-lock limitations remain disclosed. Unknown deadline/owners affect scheduling; unavailable model access prevents measuring LLM usefulness. Constant-Buy classifier output prevents a predictive-skill claim. Prototype profile-to-blend assumptions require review and do not establish suitability.

9.10 Security and Privacy Considerations

The local demo serves historical decisions on loopback and has no trading/account access. Profile persistence, production authentication and deployment security are beyond this increment.

9.11 Current Limitations

Physical projector review, remote GitHub CI, real LLM evaluation, personalised classifier modelling and deployment remain pending. The current browser journey and compiled report/deck layouts have separate review receipts. New science and the original social-classifier comparison are recorded separately. The risk comparator remains an unvalidated prototype. Institutional metadata, signatures, genuine team review and remaining research are incomplete.

9.12 Teacher Pilot Evidence

The six recorded examples, four-training/two-validation split, serving-identity limitations and untrained Phi-3-mini student are documented once in Section 6.11. No training efficacy or human-comprehension result is inferred.

Chapter 10

Remaining FYP Plan

10.1 Remaining Scope

The full approved advisory scope exceeds the current historical news-only spike. Remaining work includes social transfer/divergence, genuine risk conditioning, multi-equity evaluation, usable calibration/decision behaviour, serving parity, fault handling, local language-model evaluation/adaptation, deployment and participant validation. Transaction execution remains excluded.

10.2 Remaining Modules and Features

Restore the full runtime/data, establish the original Phi-3-mini student checkpoint, evaluate raw and emitted generations, and reconcile capacity/tolerance/lambda specifications. The paired original-candidate inference on 24 frozen held-out posts is complete; agree adequacy criteria and review the result before deployment selection. The 117 usable references from 200 acquired posts remain explicitly AI-generated. Independent human review remains optional and unmeasured. Use the declared news-only branch only with an accurately documented comparison/deviation decision; unavailable inference does not establish that neither candidate transfers. Extend beyond NVDA after acquisition/alignment assumptions and evidence are reproducible.

10.3 Remaining Iterations

The next increment validates model/runtime access and the constrained pilot. Subsequent increments integrate risk and degraded operation, social evidence/divergence and broader model evaluation, then serving/deployment and participant acceptance. QLoRA

and teacher/student work retain their later schedule; the panel feedback does not automatically make them mid deliverables.

10.4 Implementation Plan

Use the frozen explainer selection and deterministic comparators before modifying its prompt on evaluation records. Record checkpoints, model identity, data hashes and failed attempts. Restore raw/store inputs through documented acquisition or approved copies. Team-agreed owners and actual review/implementation contributions must be recorded rather than inferred from historical allocations.

10.5 Testing and Validation Plan

Run the scientific suite with documented inputs and locked dependencies, execute actual Streamlit/browser checks and GitHub CI, and assess prediction/calibration together with class distribution. For explainers, report first-attempt errors, retry/fallback rates, supported-claim/definition coverage, and human comprehension separately. The social pilot uses explicitly declared AI reference labels with prompt/input/raw-output/run provenance; candidate scores measure agreement with those references. Independent human validation is unmeasured, and optional independent review can strengthen it. Later validation covers applicable runtime requirements, fault behaviour, serving parity and participant tasks.

10.6 Deployment Plan

A delivery target remains unselected. Prepare a reproducible service and installation/deployment evidence after clean-host validation. The loopback demo and a workflow file alone do not constitute production deployment or successful continuous delivery.

10.7 Tentative Timeline

No assessment deadline has been confirmed. The sequence is dependency-based: access/restoration; explainer pilot; risk/social integration; broader evaluation; serving/deployment; acceptance and handover. Calendar dates and student ownership require team/supervisor agreement. The initial 45–75 person-hour mid-package estimate included completed work and is not a new remaining-time estimate.

10.8 Expected Final Deliverables

Reproducible source and data acquisition, model and evaluation evidence, audited decision and explanation paths, tested interfaces and degraded states, deployment/handover instructions, actual contribution/meeting records, final report/presentation and participant acceptance evidence. The staleness-gate design remains a later FYP-2 item unless scope is explicitly changed.

Chapter 11

Conclusion

Investify has a substantive historical pipeline and a tested explanation increment. Complete technical/news evidence display, independent prose/JSON checking, hard contradiction rejection and verified fallback improve the proposal-stage demonstration. The local LLM adapter and frozen pilot protocol are implemented with explicit no-model behaviour, exact maintained definitions and fair deterministic comparators. They do not establish the quality or usefulness of an unconnected model.

The preserved historical evidence remains separate from completed fresh runs. A real Alpaca experiment executes all six leakage arms at both horizons, both corrected calibration modes and retained-model/native-TreeSHAP replay. The deliberately centred canary fires, but the clean five-day score is below its random comparator and both calibrated modes remain all Buy. Lower Brier than the raw model does not establish useful decisions or superiority to a simple class-prior forecast.

Original CryptoBERT and Cardiff each complete 24 held-out MSFT posts, with macro-F1 agreement 0.2564 and 0.6016 against frozen AI references. Cardiff is the exploratory benchmark leader; the small reference, two negative examples, public pretraining overlap and lack of independent human validation prevent a broad accuracy or deployment claim. The recorded Social Study viewer validates both raw output/receipt bundles. Final selection criteria remain unset.

Continuation should complete institutional report/deck inspection, actual metadata and signatures, member contributions and genuine review/rehearsal, while preserving weak findings and original evidence. Live LLM usefulness, student adaptation, broader generalisation, risk suitability and deployment remain unmeasured. No historical ownership or participant evidence is manufactured.

The current assistant teacher construction pilot records six actual verified explanations with four training and two validation examples; the planned Phi-3-mini student remains untrained. Exact teacher serving revision is unavailable, factual JSON supervision is

assembled from original records, and no fine-tuning efficacy or human-understanding result is claimed.

Appendix A

Proposal Action Register

The exact supplied panel wording and the original proposed response register are preserved in the source snapshot. The table below records current working evidence; it does not declare supervisor-approved closure or invent owners/deadlines.

ID	Requested clarification	Current evidence and remaining closure
PF-01	Apparent contribution	Problem/contribution and research comparison drafted. Novelty/closest-work assessment and team review pending.
PF-02	LLM's user benefit	Adapter, retries, fallback and comparators implemented; six actual teacher examples replayed. Live inference, added explanatory content and comprehension benefit remain unestablished.
PF-03	Technical/sentiment explanation	All feature context and exact maintained definitions checked. No social observations or unrestricted semantic verification claimed.
PF-04	Refined problem statement	Revised wording in introduction and contribution material. Team/supervisor agreement pending.
PF-05	Pretrained models	Inventory separates FinBERT, candidate social classifiers, the current assistant teacher and planned Phi-3-mini student. Exact teacher serving revision is unavailable; exact student checkpoint and student inference remain pending.

PF-06	Explicit metrics	Prediction, calibration, explanation, social, user and system matrix and current scoped checks. Unavailable measurements labelled.
PF-07	Improved PoC beyond leakage	Corrected verifier, actual browser journey, corruption/fallback checks and versioned evidence. Supervisor demonstration and rehearsal pending.
PF-08	User transparency	Historical/constant-Buy/social/risk limits, rejected reports and labelled fallback. Full journey and human comprehension pending.
PF-09	Stock-market understanding	Referenced domain/research notes and project design implications. Corporate-action/macro/regime study and member defence review remain.

Appendix B

Detailed Requirements

Requirement	Component	Evidence
MID-FR-01	Schema/catalogue/input handling	Actual-record and missing/invalid-input checks
MID-FR-02	Context and both interfaces	All 372 records; complete coverage tests
MID-FR-03	Verifier	Prose/payload contradiction regressions
MID-FR-04	Verifier/emission	Hard gates, omissions, malformed claims, verified fallback
MID-FR-05	Emission/reporting	Retained failed report and HTTP assertions
MID-FR-06	Strict prose gate	Clean acceptance and unsupported-sentence rejection
MID-FR-07	Structural readiness/UI control	Missing, empty, truncated, invalid-type and read-denied fixtures; module discovery and actual widget checks
MID-FR-08	Local adapter and interfaces	Retry, unavailable-provider, malformed/truncated output, loopback transport and fallback tests
MID-FR-09	Guided template and verifier	All 372 guided records; modified-definition and coverage checks
MID-FR-10	Evaluation command	Disjoint chronological selection, plan-only and zero-completion tests

MID-FR-11	Complete guided contract	Non-top-k, missing/zero, family, target, scope and two-source coverage tests
MID-FR-12	Prototype profile/form	All 32,400 answer combinations; cap, incomplete-answer and actual widget tests
MID-FR-13	Retrospective scaling	Both 148/38 splits, held-out perturbation and development/missing-input refusal
MID-FR-14	Separate comparator	Actual HTTP/AppTest independence; 228 real-row/labelled-scenario arithmetic checks
MID-FR-15	Social-study provenance/plan	Fixture provenance checks plus the actual frozen 200-post/117-reference AI study
MID-FR-16	Sentiment comparison tool	Matched-plan, per-model coverage/refusal, null partial metrics and explicit ties/raw-output tests; actual paired 24-post scores recorded
MID-FR-17	Offline classifier runner	Eight backend-fixture checks; head/mapping/weight refusal, no label inputs, raw/error/truncation retention; both pinned original classifiers executed
MID-FR-18	Recorded teacher import/export	Seven fixture tests plus actual evidence replay and viewer checks; six real examples, hybrid target disclosure and separate exports

Mapping to the complete proposal baseline and final acceptance study remains pending. Exact functions and named passing tests are retained in `code/docs/requirements_traceability.json`; `tools/verify_requirements_traceability.py` checks their existence and matching selected-suite execution. The working Markdown matrix also distinguishes each original first-mid commitment from downstream saved-record engineering evidence.

Appendix C

Detailed Test Cases

The table records the earlier 12 October selected engineering scope. Later fresh science, original-classifier and guided-demo receipts below supersede its pending states; scopes overlap and are not summed.

Group	Scope	Status
Existing verifier/emission	20 checks, including clean fixtures and corruptions	Passed
Context	5 tests; actual records, absent/missing features	Passed
Verifier regressions	11 tests; contradictions, omissions, schema, padding, strict mode, report retention	Passed
Portable HTTP	11 tests; verification/fallback, invalid inputs, profile contract and independent risk comparator	Passed
Portable request ownership	12 controlled Node cases execute real inline handlers with limited DOM/deferred-fetch fixtures; separate eight pure-state assertions	Passed; browser QA pending
Harness structural readiness	10 tests; missing/empty/truncated/wrong-type/read-denied files, module discovery and explicit preflight limits	Passed
Local generator	12 tests; definitions, positive acceptance, retries/fallback, malformed/truncated output, configuration and provenance	Passed
Provider transport	2 actual loopback HTTP fixture checks	Passed

Evaluation protocol	7 checks; chronological split, frozen plan enforcement, no-inference/zero-output accounting	Passed
Complete guided evidence	8 tests; all-feature coverage, non-top-k corruptions, missing/zero, family and scope checks	Passed
Risk profile/comparator	10 tests; 32,400 answer combinations, chronology, scaling isolation, failure and classifier independence	Passed
Authoritative record validation	4 tests; non-finite fields, vector/rank/value consistency and invalid identity/horizon	Passed
Social-selection tooling	9 labelled-fixture tests; human-label provenance schema, adjudication, fixed split, paired coverage and metric/refusal paths	Passed; actual study pending
Source revision/handover	5 tests; checked Git-free inventory, explicit working-tree status, modified/unsafe inventory refusal and ancestor-repository isolation	Passed
Offline sentiment inference	8 labelled-backend tests; frozen identity, text/label mapping, head refusal, failures, truncation and absent weights	Passed; actual model pending
AI annotation/import	10 labelled-fixture tests; blinded packets, retained initial labels, null single-pass reliability, input/provenance/persistence checks	Passed; human validation pending
Generative sentiment protocol	6 labelled-provider tests; frozen parent split/runtime, raw failures and identity/source-drift refusal	Passed; actual inference pending
Original-publisher gates	4 labelled-file tests; exact catalog identity, complete shard hashes and receipt replay	Passed; verified real weights pending
Recorded teacher import/export	7 fixture tests; date grouping, actual-versus-fixture gate, independent prose checks and separate exports	Passed; student pending

Streamlit AppTest	20 tests; existing flow plus explicit profile, real comparator and stale/development refusal	Passed
Live endpoint exercise	62 checks, including explanation cases and separate profile/refusal probes	Passed
Comparator arithmetic audit	228 checks on 76 held-out real rows and three labelled parameter scenarios	Passed

The selected suite totals 212 passes and zero skips. Test XML, fixture confusion matrix, artifact/source validation, comparator audit and live HTTP evidence are in `Investify/evidence`. Streamlit cases ran in the isolated restored viewer environment; remote CI has not run. Scenario profiles are deterministic probes, not participant observations. The dated initial continuation receipt is `background_readiness_tests_20261012.xml`; the later 188-pass receipt is `background_delivery_tests_20261012_v2.xml`. The final current receipt and source bundle identify any subsequent matching snapshot. The separate Node handler receipt is `background_portable_handlers_20261012_parallel.json`; its twelve cases are not added to Python-suite totals and perform no live server/model/browser requests. The 18 requirement IDs have exact source/test anchors checked by the traceability tool. This establishes execution of named regressions, not complete scientific or institutional acceptance.

The matching post-recovery receipts are `recovery_selected_tests_20261012.xml` (212 passes, twenty Streamlit cases) and `recovery_core_tests_20261012.xml` (153 scoped passes, three deselected). These overlap and are not added together. Separate read-only Python 3.13 probes execute three unchanged shuffled-split function bodies with real sklearn and replay actual isotonic helper fits on explicitly labelled numeric fixtures. Neither probe restores the original scientific lock or uses historical market inputs.

The 13 October guided-demo receipt records 28 passes and zero skips, including 24 AppTest cases and four recorded-study selector checks. It verifies fresh defaults, explicit fictional profiles, profile reset, navigation state and failure/fallback. The 179-pass scientific contract scope retains six legacy-input skips, with eight separate admission-refusal passes. Both original classifiers complete the same 24 held-out AI references. Chapters 6 and 7 document these actual studies; browser review, human validation and remote CI remain unexecuted.

Appendix D

Repository and Reproduction

D.1 Active Workspace

Work in `E:/Uni_Stuff/FYP/Investify`; run Python commands from its `code` directory. The local branch is `main`, with no configured remote. The original checkout and source archives remain separate. Local commit `93d0d60` records the fresh scientific checkpoint; subsequent source and artifact changes retain dated receipts.

D.2 Guided Demo and Historical-Record Demonstration

The current Streamlit app opens with Overview and renders one workspace at a time. Fresh corrected records are the default; its Walkthrough has four distinct steps. Run from the code directory:

```
python -B -m streamlit run app/streamlit_app.py ‘
  --server.address=127.0.0.1 --server.port=8507 ‘
  --server.headless=true
```

Open the printed loopback URL. Fictional profile examples are explicit user actions; no participant data is assumed. The original portable interface remains available separately below. Use the restored viewer environment or the code's portable Python path:

```
python -B scripts/serve_mideval_demo.py --port 8768
```

Open the printed loopback address. The demonstration validates historical records and explanations; it does not fetch prices, retrain models or generate unconfigured LLM output.

D.3 Executed Fresh Scientific Snapshot

The separately identified snapshot lives under `code/artifacts/scientific_runs/alpaca_revision_conservative_20261012_run01`. It retains configuration, a new price store, corpus manifest, all-arm results, both calibration-mode records/traces and execution logs. FinBERT scores are cached on E: under the same run identity. Original September bytes and the exact scientific lock are not reproduced.

Use `code/scripts/prepare_fresh_alpaca_science.py` for new receipt-verified input preparation, `code/scripts/prepare_scientific_recovery.py` for preflight/execution and `code/scripts/verify_fresh_science_run.py` for actual model/calibration/native-TreeSHAP replay. Each new execution must use new output/cache/receipt identities. The current scientific Python 3.11 environment is separate from the historical viewer lock.

D.4 Original Social-Classifiers

The original V3 benchmark, label freeze and chronology remain unchanged. `code/artifacts/mid_eval/social_study_20261013_run02` contains the new source-bound plan, both original-classifier bundles and their recomputed comparison. Its complete publisher catalog is retained in the preceding preparation directory. Weights and the disclosed lossless Cardiff serialization remain on E: caches. The recorded viewer uses `social_latest_study.json`; it recomputes receipt-bound agreement scores rather than inferring new predictions.

D.5 Report and Handover Limits

The supplied institutional class and logo retain their original bytes. Main-source compatibility settings allow the portable Tectonic engine to compile the draft without changing that class. Compiler/cache/build identities and layout checks are recorded separately. Required institutional facts, applicable signatures and actual team review remain open. Existing ZIPs and receipts prove their own dated snapshots; no remote CI, deployment, student ownership or final submission is inferred from local checks.

Appendix E

Additional Design Diagrams

The implemented architecture appears in Chapter 5. The following diagram describes the current runtime/JSON objects, not a deployed relational database. Saved records are inputs; explanations, verification and profile-comparator results are transient. A full-product database design remains pending.

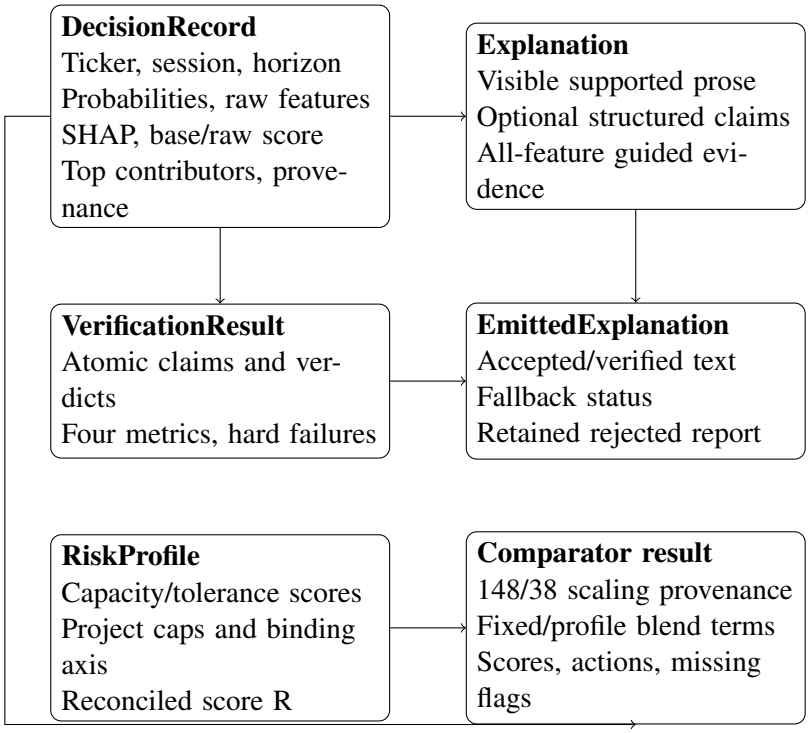


Figure E.1: Implemented record, explanation and profile-comparator object relationships.

Appendix F

Meeting and Progress Evidence

Actual meeting dates, participants, supervisor feedback and agreed actions have not been supplied for this increment. No attendance, approval or member contribution is inferred. The team must add authentic records before submission, keeping dates/participants/actions traceable to their original evidence.

Appendix G

Handbook Mid-1 Compliance and Document Control

G.1 Source and Stage Interpretation

The supplied 2026 draft handbook Table 1a (PDF p.11), stage intent (p.27), Forms 2/3 (pp.37–38), report contents (p.23) and baseline mapping (pp.56–59) are reviewed with the supplied four-page cohort Mid-Evaluation rubric. The screenshot directive matches Table 1a. Mid-1 is a week-eight, supervisor-only formative checkpoint with a recommended fifteen-minute presentation and twenty percent of FYP-1. The forty-five percent implementation expectation belongs to Final-1. The draft explicitly makes cohort stage requirements authoritative over its proposed baseline sequencing.

G.2 Document Control

Field	Value
Project	Investify, F26-003-Investify
Document	FYP-1 Mid Report, version 1.0 working revision 3
Preparation	13 October 2026; assessment date unconfirmed
Authorship disclosure	Assistant-generated code/design/report drafts and verification; actual student review/contributions require genuine records.
Status	Draft for review; no supervisor/stakeholder baseline approval or signature claimed.

Change	User requested complete UI rework, elimination of empty report pages, additional project-specific models and handbook compliance audit. Scientific freezes remain unchanged.

G.3 Artifact Coverage

ID	Required evidence	Present evidence and closure
H01	Report v1.0 and document control	Draft; signatures/metadata pending
H02	Project vision, scope and stakeholders	Written; stakeholder/supervisor review pending
H03	Related work, research gap and CCP	Written; genuine source defence pending
H04	Use cases and use-case diagram	Implemented paths documented
H05	SRS: numbered, prioritized FRs/NFRs	Draft baseline; acceptance matrix added
H06	Software design specification and rationale	Revised; code-linked models supplied
H07	Test plan: levels and techniques	Current unit/module/system scopes separated
H08	Development and iteration plan	Revised plan; genuine owners/dates pending
H09	Wireframes	Design wireframe present
H10	Actual UI screens and demonstrable flow	Actual Edge journey/captures in Chapter 7; projector/rehearsal pending
H11	DEI, ethics, privacy and legal statements	Written; rights/clearance sign-off not invented
H12	Iteration-1 implemented core module	Actual live checking/profile logic; tested
H13	Domain/class model	Actual classes/entities and invariants explained
H14	Component/layer/structure design	Component/layer models selected; no redundant structure chart
H15	Flow/DFD/data dictionary	Selected data-flow model and field rules present
H16	Activity and sequence/SSD	Failure/fallback path and messages present

H17	State/network/interaction behavior	State model selected for stale-check handling; graph/interaction alternatives not required
H18	DB representation/schema/ERD as applicable	Artifact schema present; no implemented relational DB claimed
H19	Data structures and algorithms	Contracts and steps documented
H20	Comments and naming conventions	Snake-case functions and documented invariants
H21	Static code analysis	Six current files pass scoped Ruff E4/E9/F checks
H22	Unit/module tests and test cases	Executed scopes retained; overlap not summed
H23	CI/CD and version-control process	Workflow exists; remote CI/CD and member evidence pending
H24	Deployment diagram	Actual localhost topology documented
H25	System-level validation/testing	Executed local tests; no external-user acceptance claimed
H26	Configuration/setup and tool manual	Current launch and failure recovery documented
H27	Working code plus live demonstration	Live arithmetic/verifier; recorded ML/study outcomes distinguished
H28	Response to proposal actions	Evidence linked; genuine closure remains pending
H29	Design evolution since proposal	Earlier UI and verifier choices compared with current code
H30	Code walkthrough and genuine member ownership	Walkthrough prepared; actual member defence not observed
H31	Milestone, delay and recovery record	Technical blockers recorded; owners/dates require team input
H32	Requirements/design baseline approval	Draft only; no stakeholder/supervisor approval claimed
H33	Dependency/API/license register	Register started; data-use and clearance review pending
H34	Responsible AI/tool disclosure	Assistant work disclosed; student review still required

G.4 Dependency, Access and Responsibility Register

Dependency	Current use	Rights/access and limitation
FinBERT, XGBoost, TreeSHAP	Actual separately recorded scientific run	Publisher revisions/bytes and numerical replay verified; useful prediction remains unestablished.
CryptoBERT and Cardiff	Original 24-post classifier comparison	Original weights retained; AI-reference agreement remains exploratory; deployment adequacy not approved.
Alpaca/Benzinga	Authenticated fresh prices/news	Access and source receipts retained; historical availability and public redistribution rights not inferred.
StockNet MSFT posts	Frozen source/reference pilot	Source declared; underlying tweet rights not authenticated by a repository code license. No public redistribution approval claimed.
Python/Streamlit and UI packages	Local prototype runtime	Installed identities and dependency sets recorded; runtime is separate from scientific lock.
Assistant/GenAI tools	Code, figures, drafts, tests and teacher pilot	Heavy assistance disclosed. Student review, modification ability and genuine defence are not inferred.
Human participants	No completed user study	No consent, comprehension benefit or ethics clearance is manufactured; future studies require applicable review.

G.5 Open Genuine Decisions

Confirm programme dates, co-supervisor applicability and coordinator. Record actual member work, review and code understanding. Baseline the scope/SRS/SSD with the real supervisor/stakeholders; close proposal actions through genuine evidence. Review data/model use and any required ethics/permission. Name owners and revised dates for

outstanding milestones. Captured browser layouts have been reviewed; physical projector review and rehearsal still require genuine observation.

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